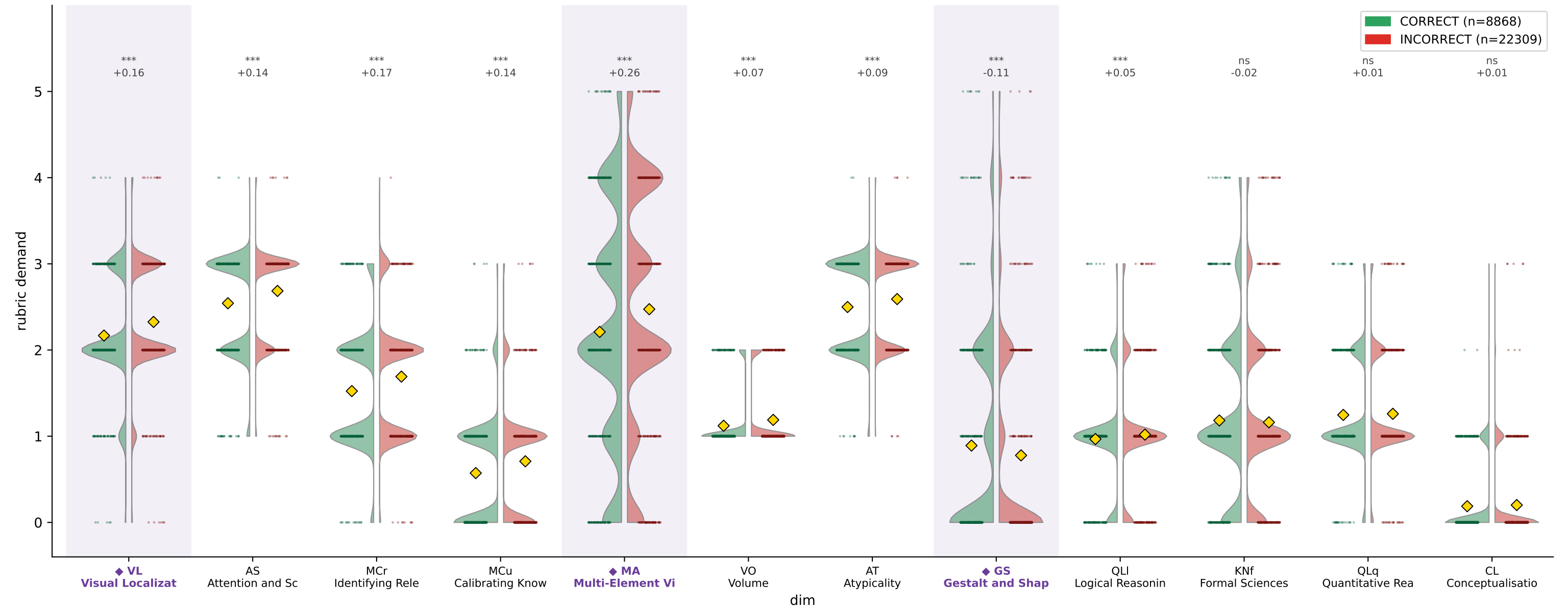
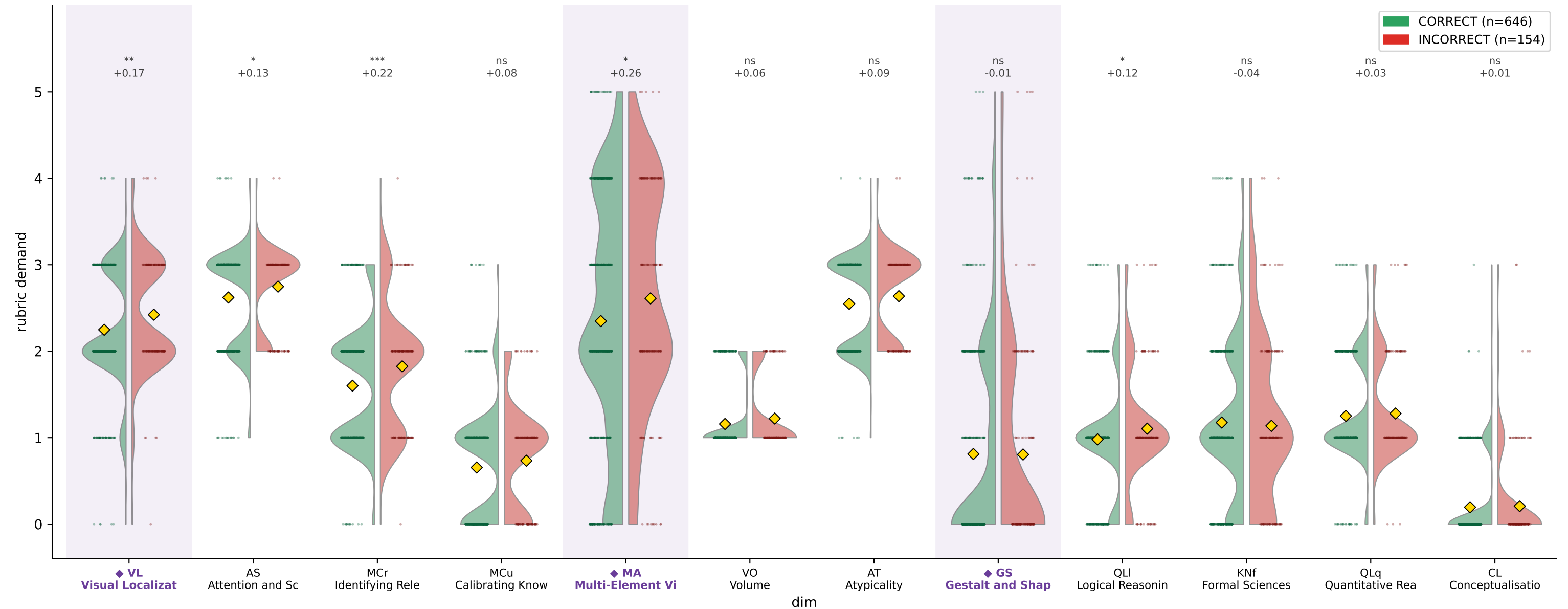


CharXiv DeLeAn demand by reasoning correctness — POOLED across 39 models (train, n=31177 obs)



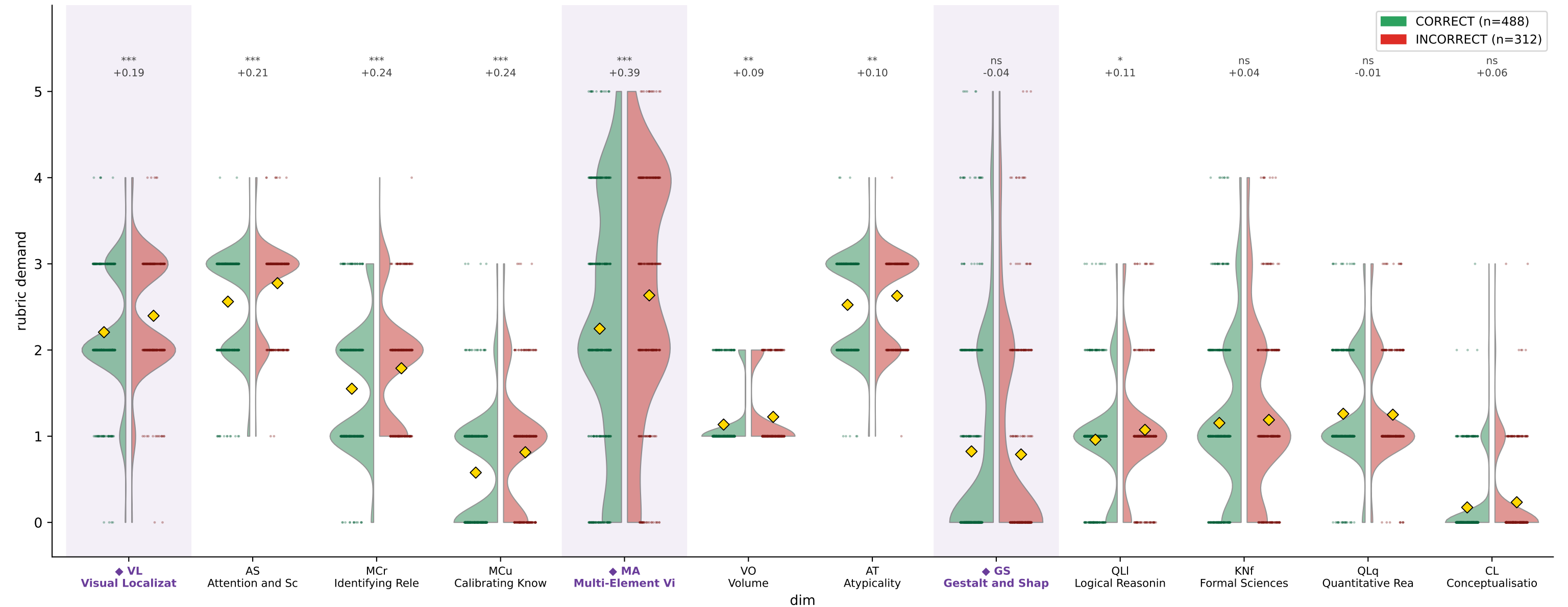
Violins = demand distribution (KDE); jittered points = items (≤ 600 /group shown); ♦ diamond = mean. Δ = incorrect – correct (positive $\Rightarrow$ failed items higher-demand). Mann-Whitney U: *** $p < .001$, ** $p < .01$, * $p < .05$, ns. Dims in fixed global order; ♦/purple = emergent

CharXiv DeLeAn demand by reasoning correctness — Human (train)

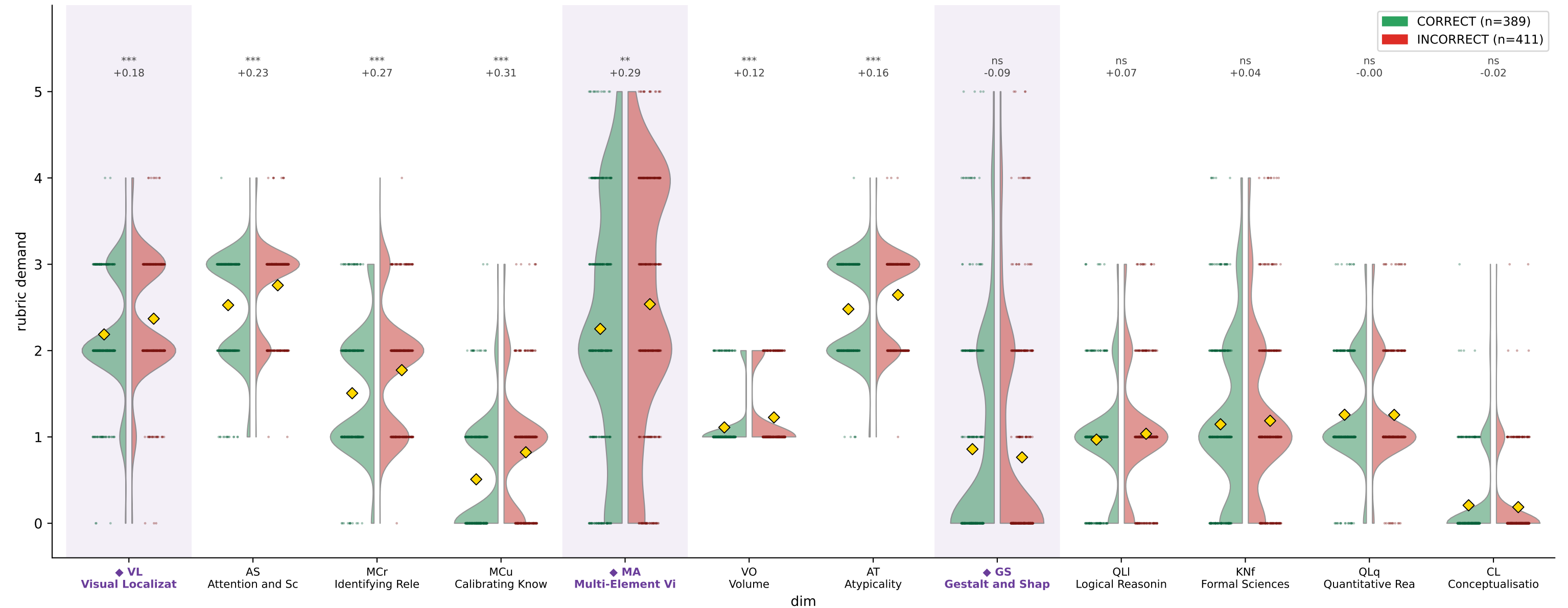


Violins = demand distribution (KDE); jittered points = items (≤ 600 /group shown); $\blacklozenge$ diamond = mean. Δ = incorrect – correct (positive $\Rightarrow$ failed items higher-demand). Mann-Whitney U: *** $p < .001$, ** $p < .01$, * $p < .05$, ns. Dims in fixed global order; $\blacklozenge$ /purple = emergent

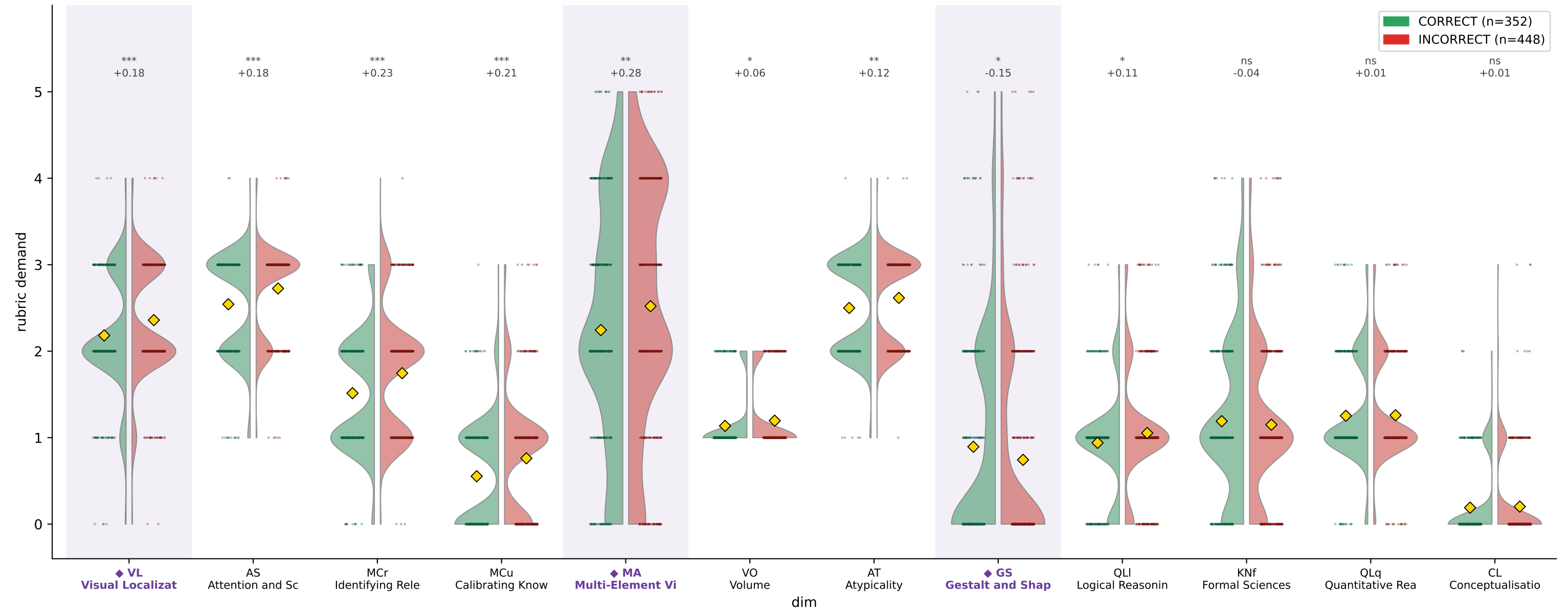
CharXiv DeLeAn demand by reasoning correctness — Claude-3-5-Sonnet (train)



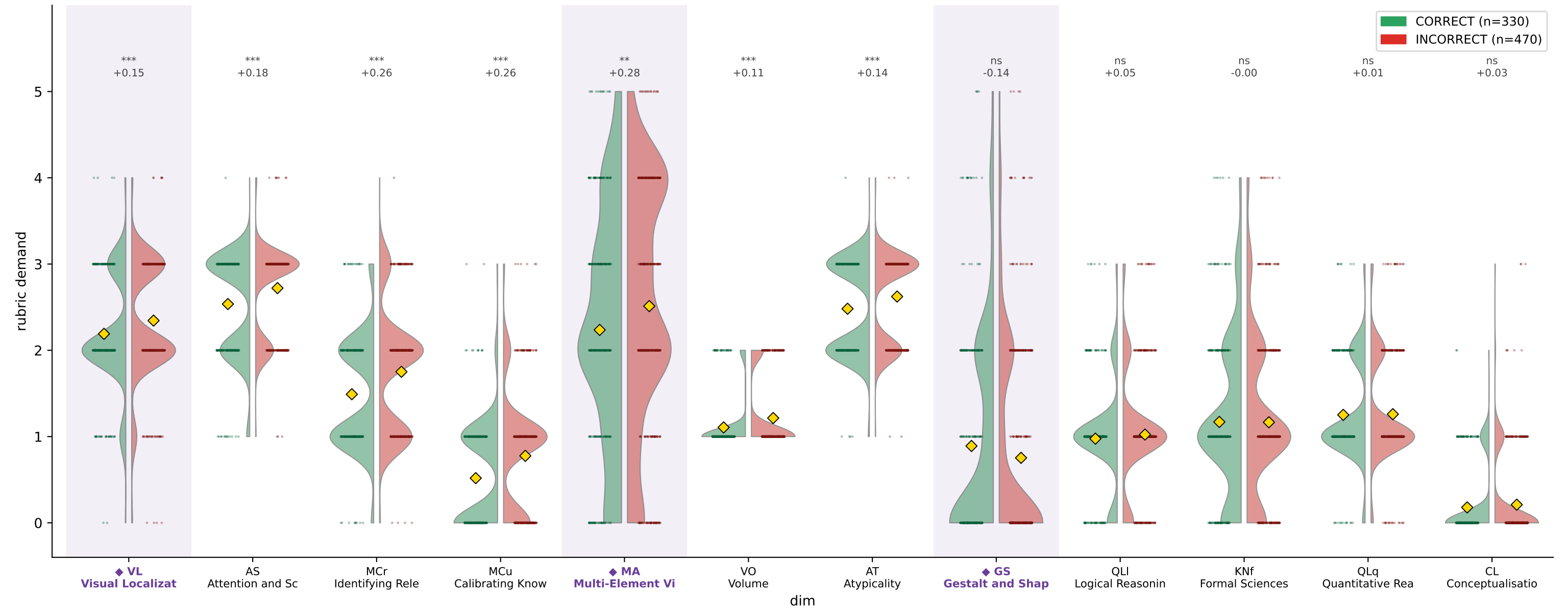
CharXiv DeLeAn demand by reasoning correctness — GPT-4o (train)



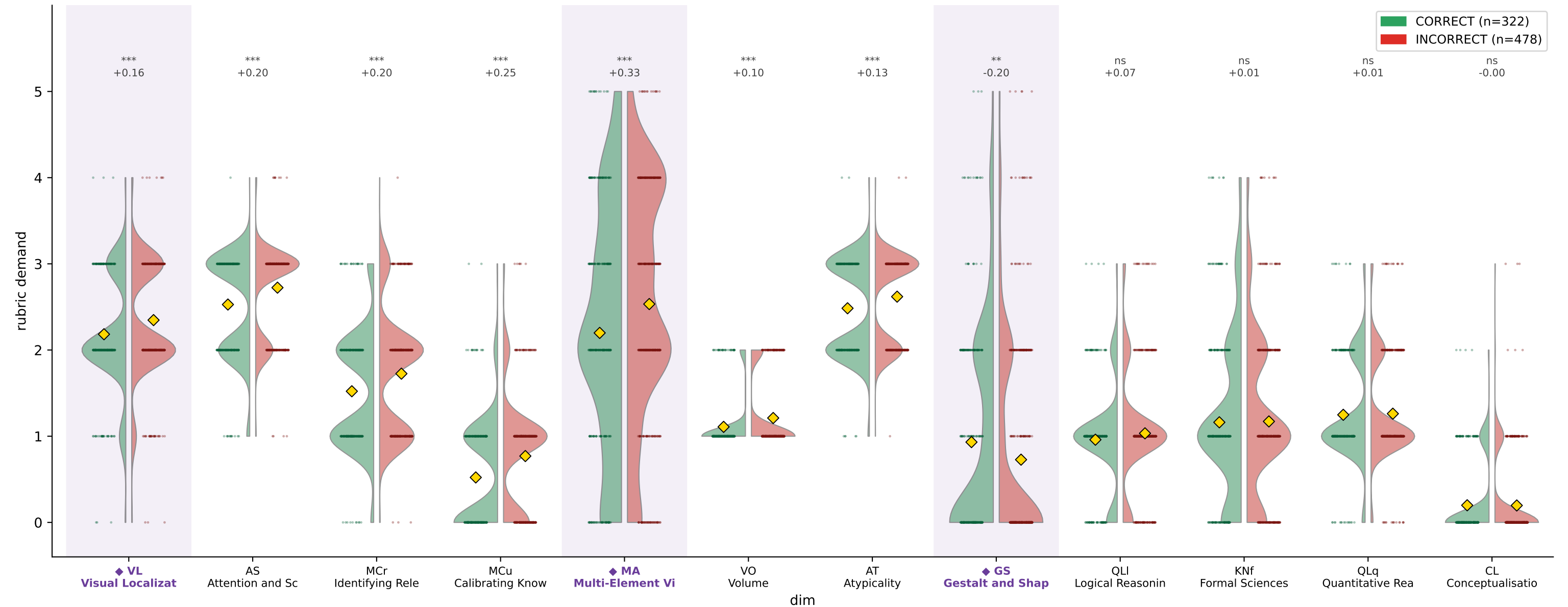
CharXiv DeLeAn demand by reasoning correctness — Gemini-1-5-Pro (train)



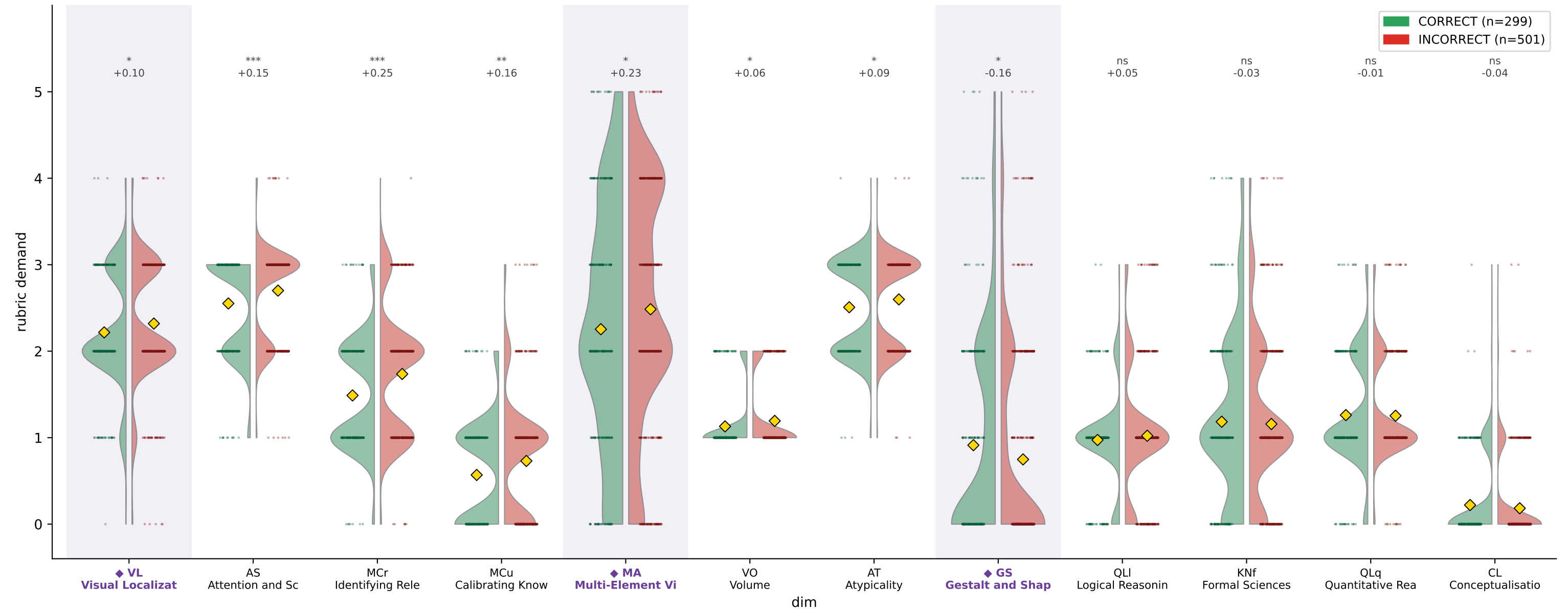
CharXiv DeLeAn demand by reasoning correctness — InternVL2-Pro (train)



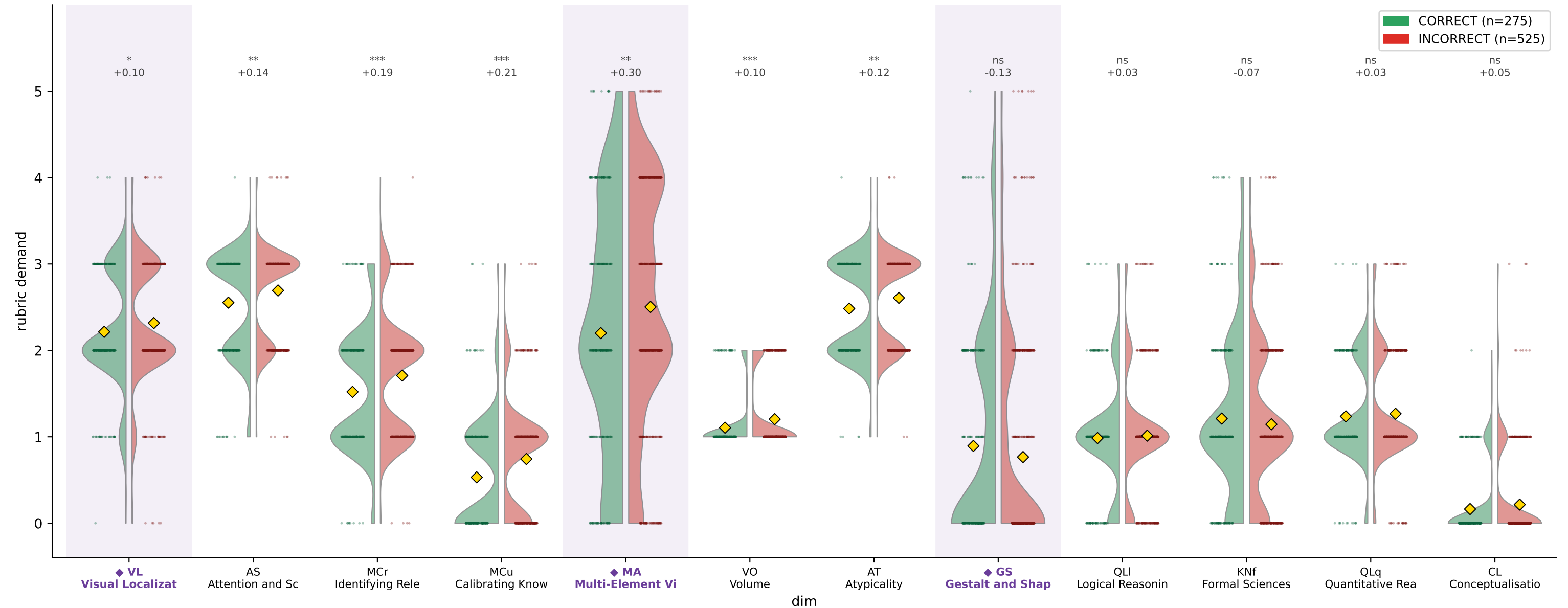
CharXiv DeLeAn demand by reasoning correctness — InternVL2-76B (train)



CharXiv DeLeAn demand by reasoning correctness — GPT-4V-Turbo (train)

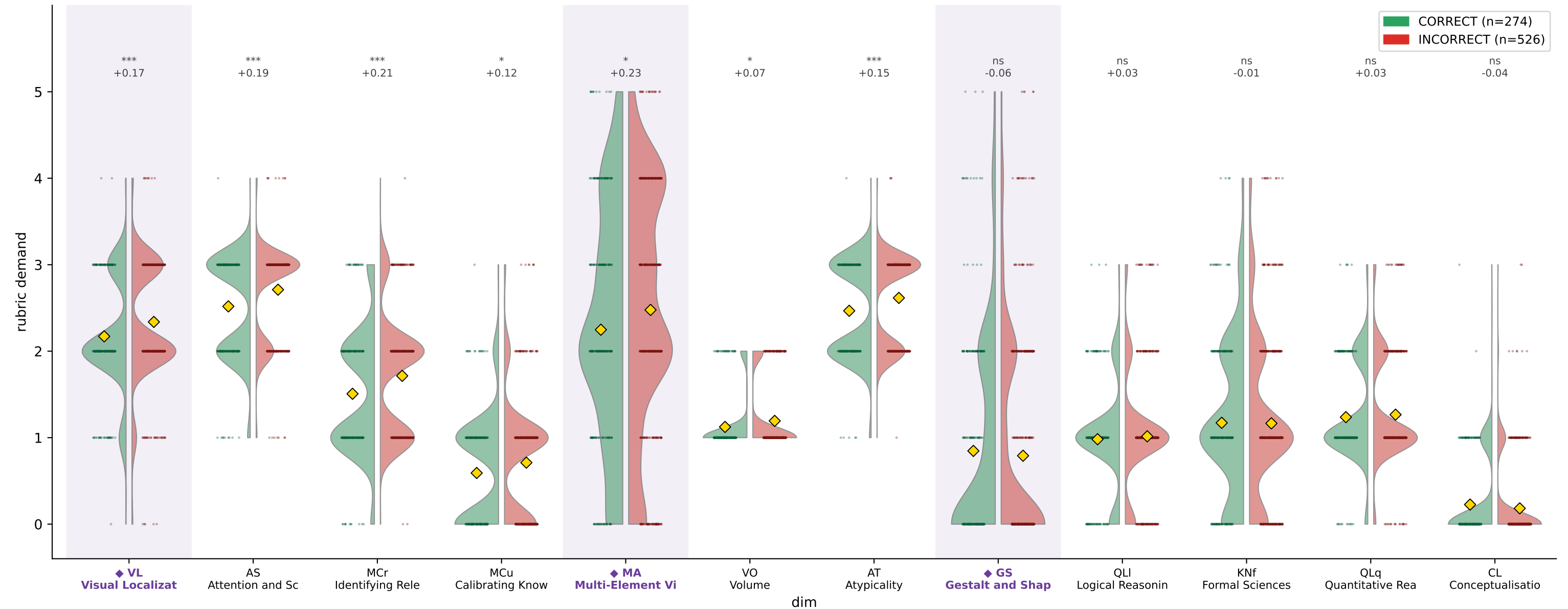


CharXiv DeLeAn demand by reasoning correctness — InternVL2-26B (train)

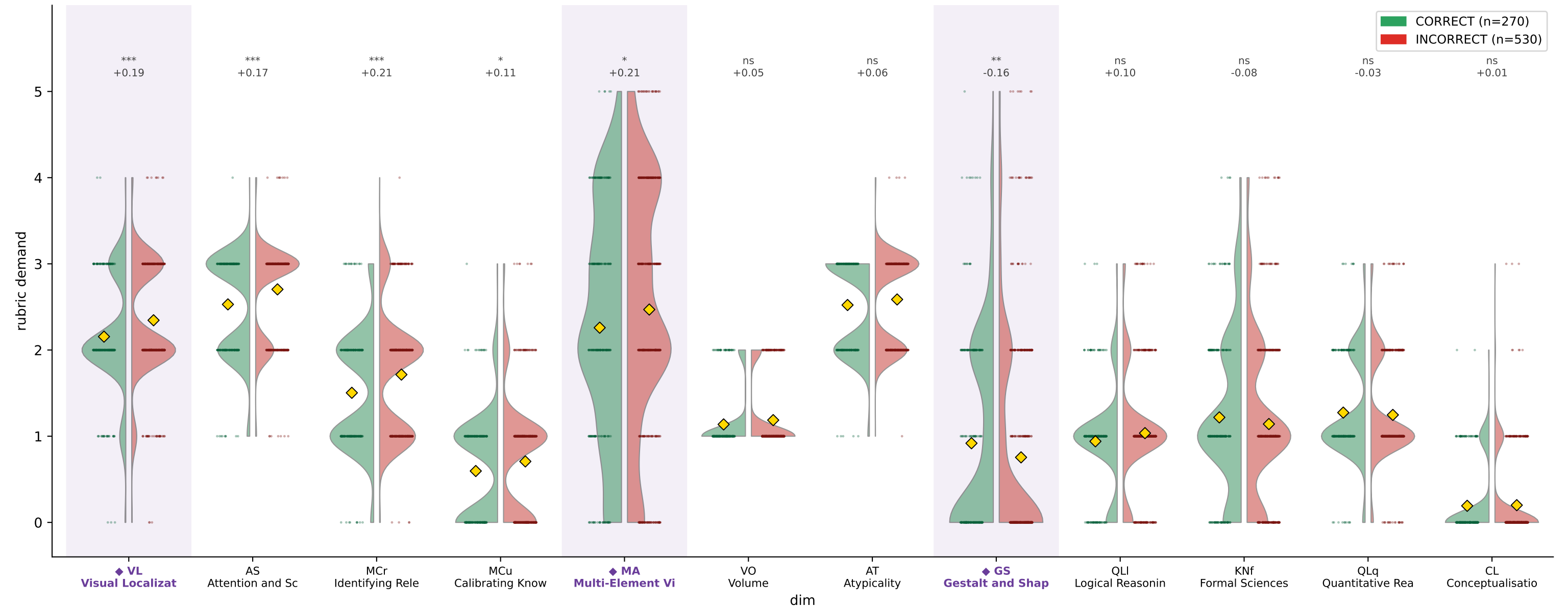


Violins = demand distribution (KDE); jittered points = items (≤ 600 /group shown); $\blacklozenge$ diamond = mean. Δ = incorrect – correct (positive $\Rightarrow$ failed items higher-demand). Mann-Whitney U: *** $p < .001$, ** $p < .01$, * $p < .05$, ns. Dims in fixed global order; $\blacklozenge$ /purple = emergent

CharXiv DeLeAn demand by reasoning correctness — GPT-4o-mini (train)

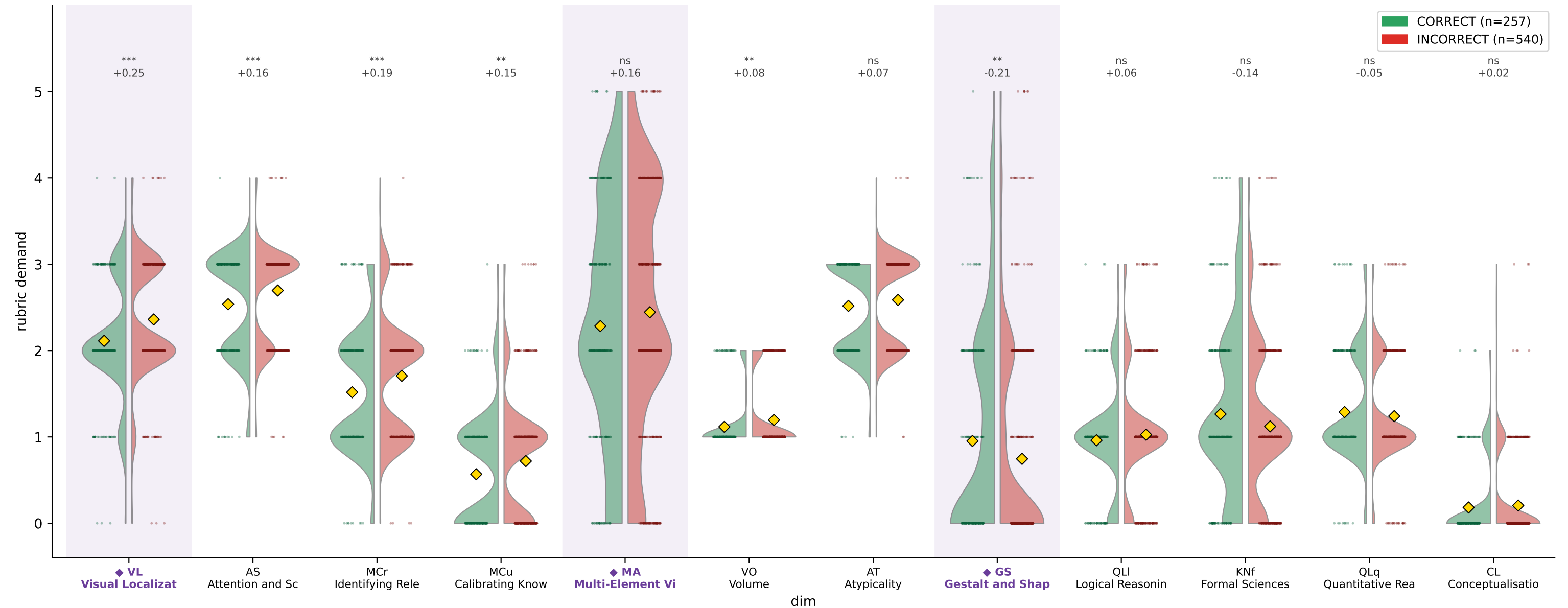


CharXiv DeLeAn demand by reasoning correctness — Claude-3-Sonnet (train)

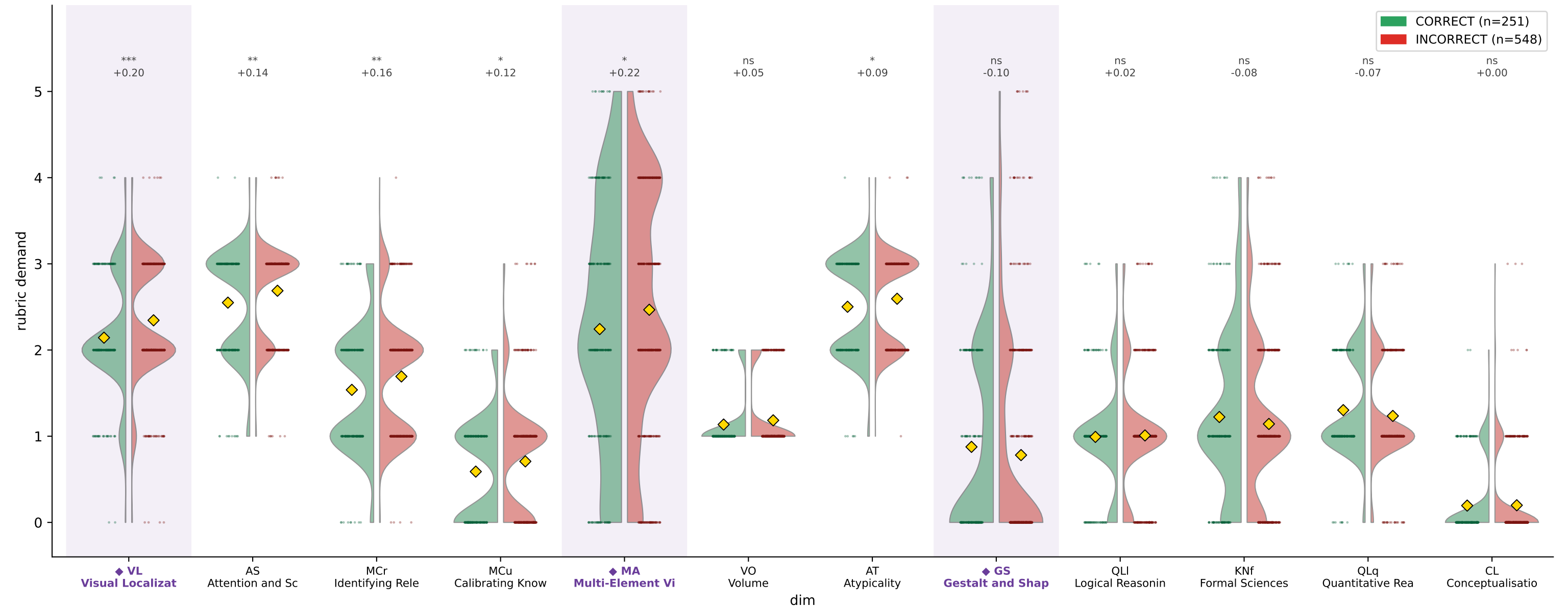


Violins = demand distribution (KDE); jittered points = items (≤ 600 /group shown); ♦ diamond = mean. Δ = incorrect - correct (positive $\Rightarrow$ failed items higher-demand). Mann-Whitney U: *** p<.001, ** p<.01, * p<.05, ns. Dims in fixed global order; ♦/purple = emergent

CharXiv DeLeAn demand by reasoning correctness — Claude-3-Haiku (train)

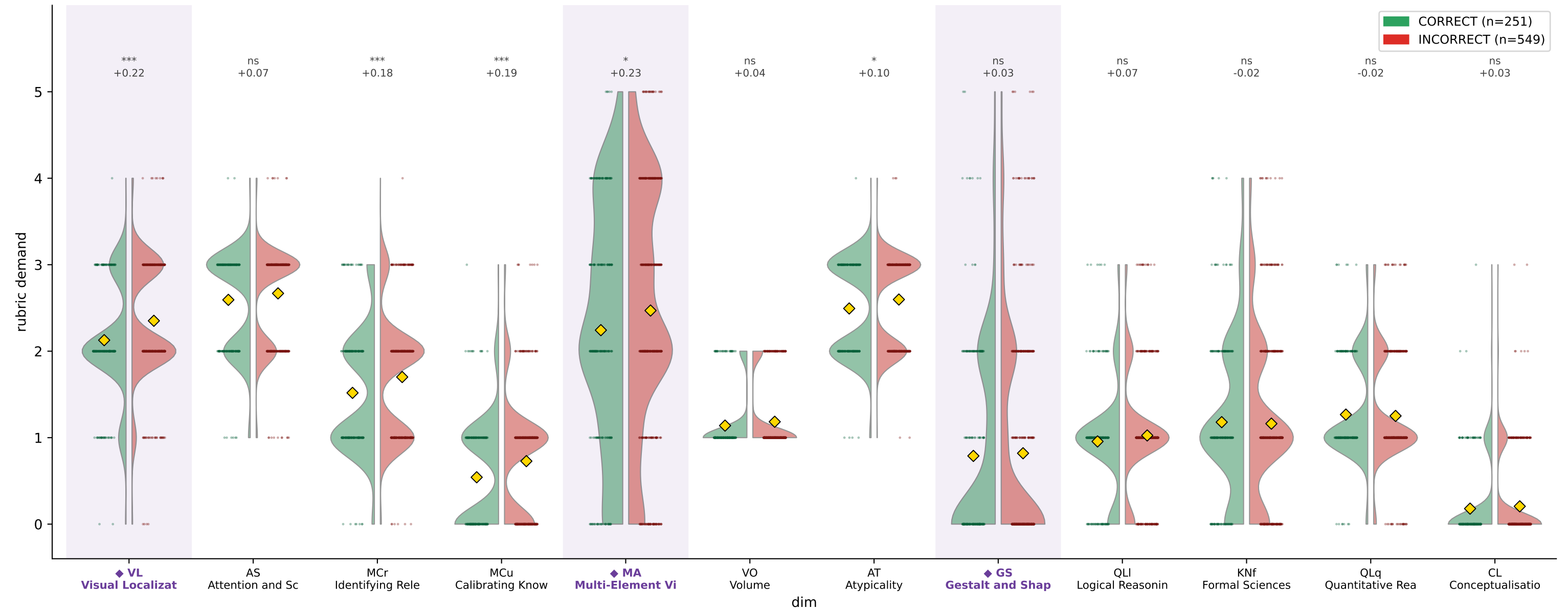


CharXiv DeLeAn demand by reasoning correctness — Claude-3-Opus (train)

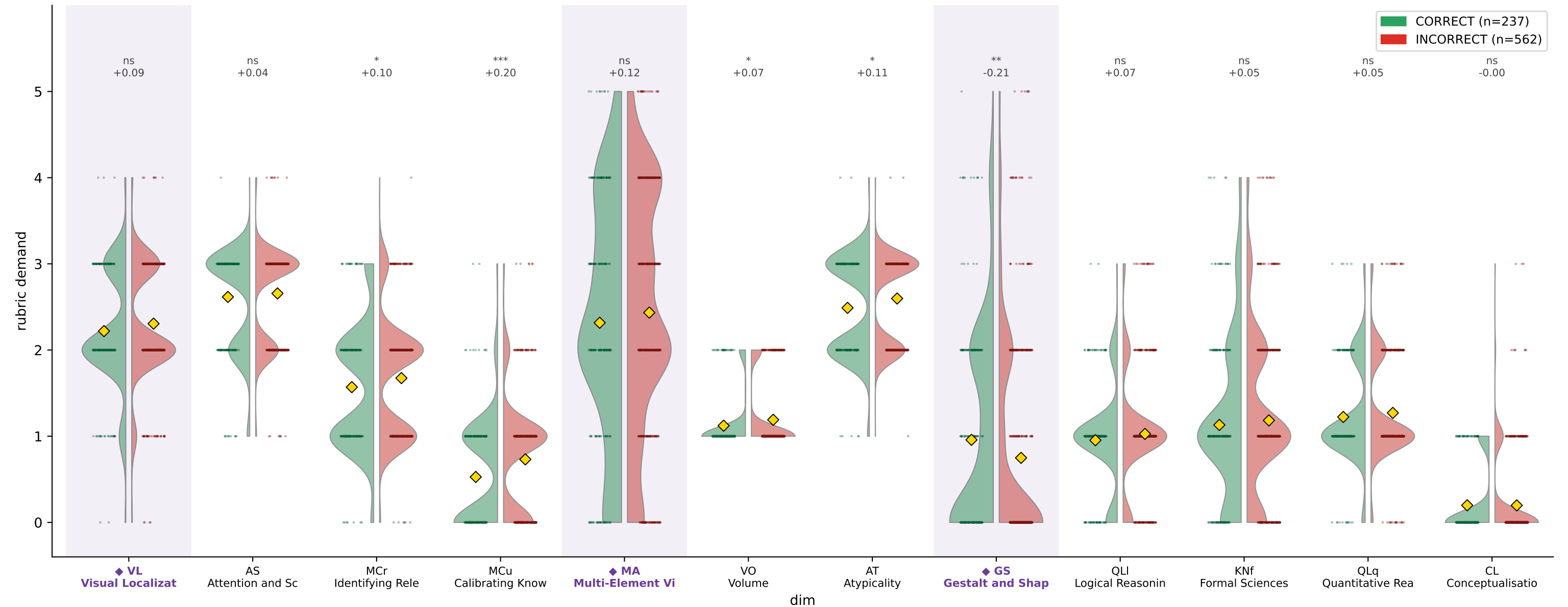


Violins = demand distribution (KDE); jittered points = items (≤ 600 /group shown); $\blacklozenge$ diamond = mean. Δ = incorrect – correct (positive $\Rightarrow$ failed items higher-demand). Mann-Whitney U: *** $p < .001$, ** $p < .01$, * $p < .05$, ns. Dims in fixed global order; $\blacklozenge$ /purple = emergent

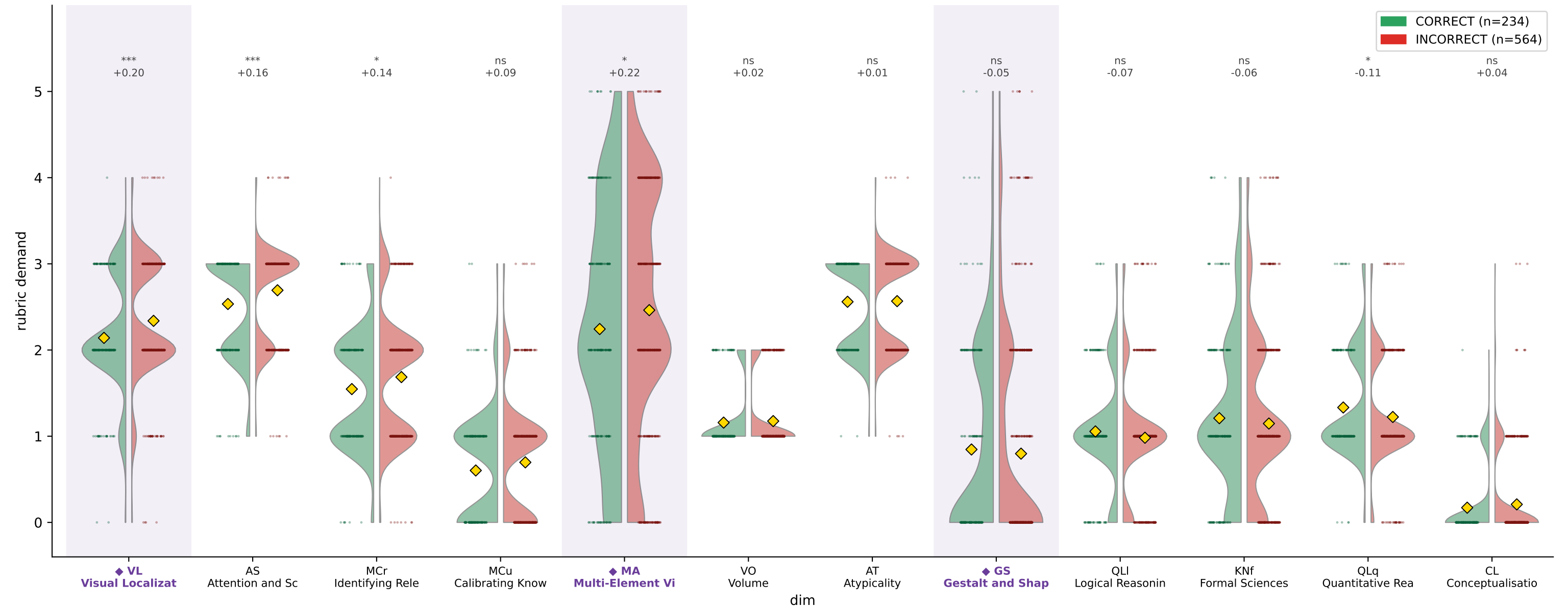
CharXiv DeLeAn demand by reasoning correctness — Phi-3-Vision (train)



CharXiv DeLeAn demand by reasoning correctness — InternVL-Chat-V1-5 (train)

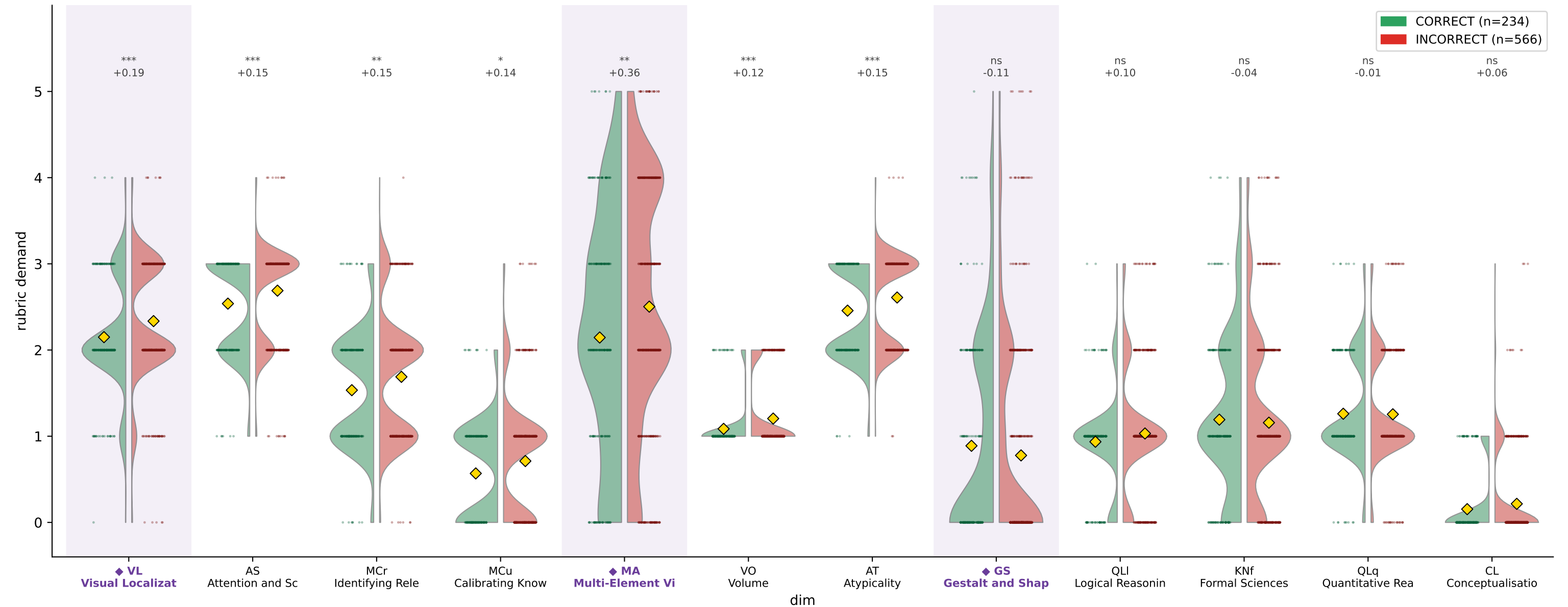


CharXiv DeLeAn demand by reasoning correctness — Reka-Core (train)



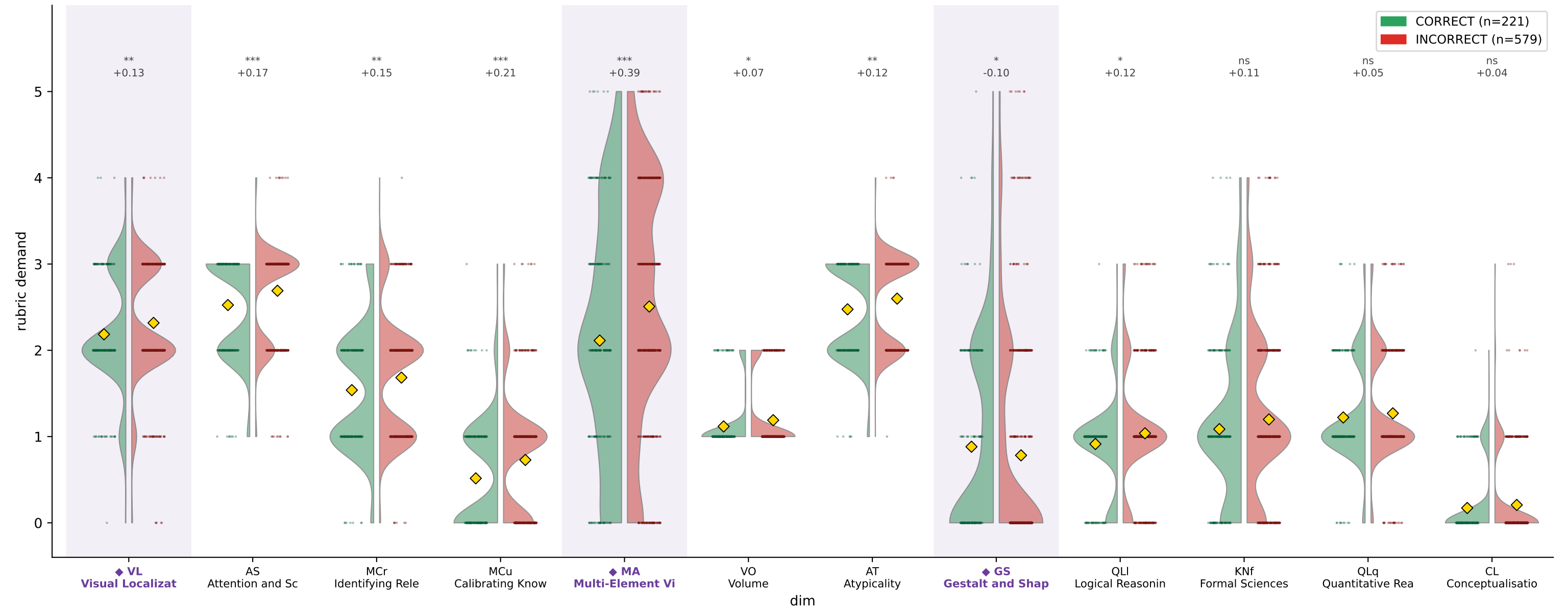
Violins = demand distribution (KDE); jittered points = items (≤ 600 /group shown); $\blacklozenge$ diamond = mean. Δ = incorrect – correct (positive $\Rightarrow$ failed items higher-demand). Mann-Whitney U: *** $p < .001$, ** $p < .01$, * $p < .05$, ns. Dims in fixed global order; $\blacklozenge$ /purple = emergent

CharXiv DeLeAn demand by reasoning correctness — Ovis1.5-Llama3-8B (train)

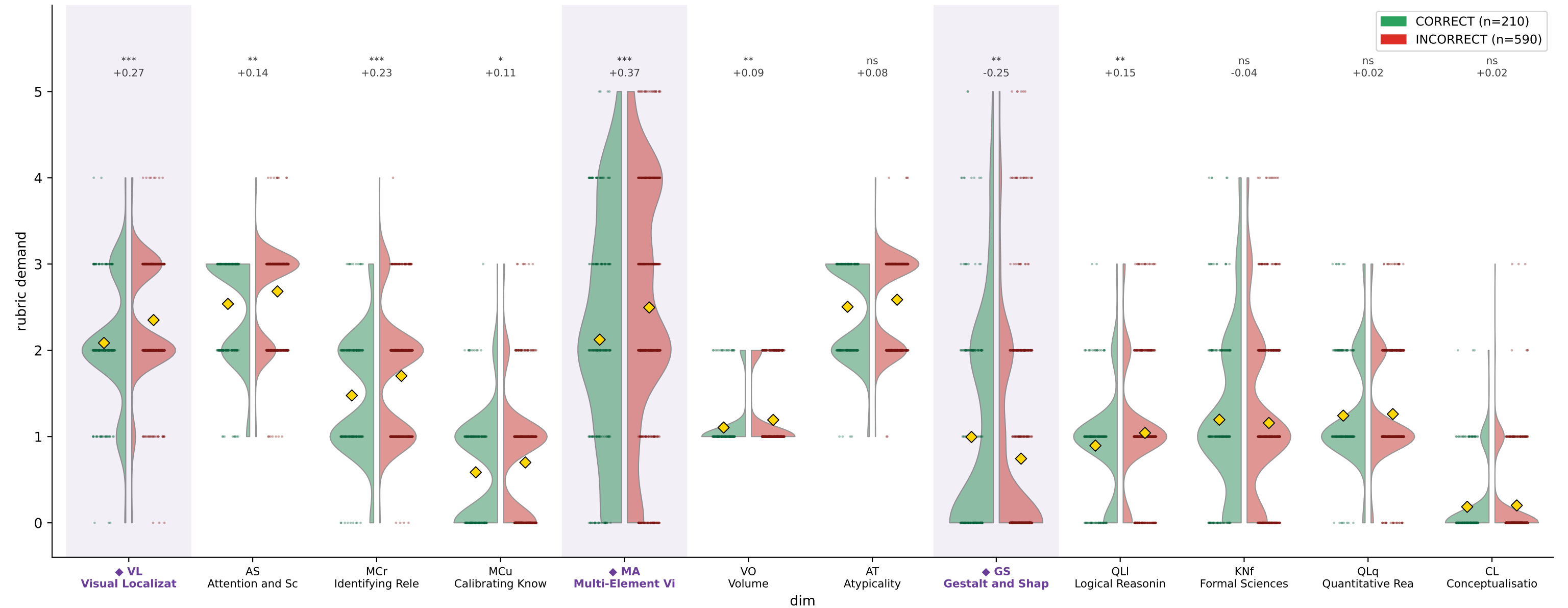


Violins = demand distribution (KDE); jittered points = items (≤ 600 /group shown); $\blacklozenge$ diamond = mean. Δ = incorrect – correct (positive $\Rightarrow$ failed items higher-demand). Mann-Whitney U: *** $p < .001$, ** $p < .01$, * $p < .05$, ns. Dims in fixed global order; $\blacklozenge$ /purple = emergent

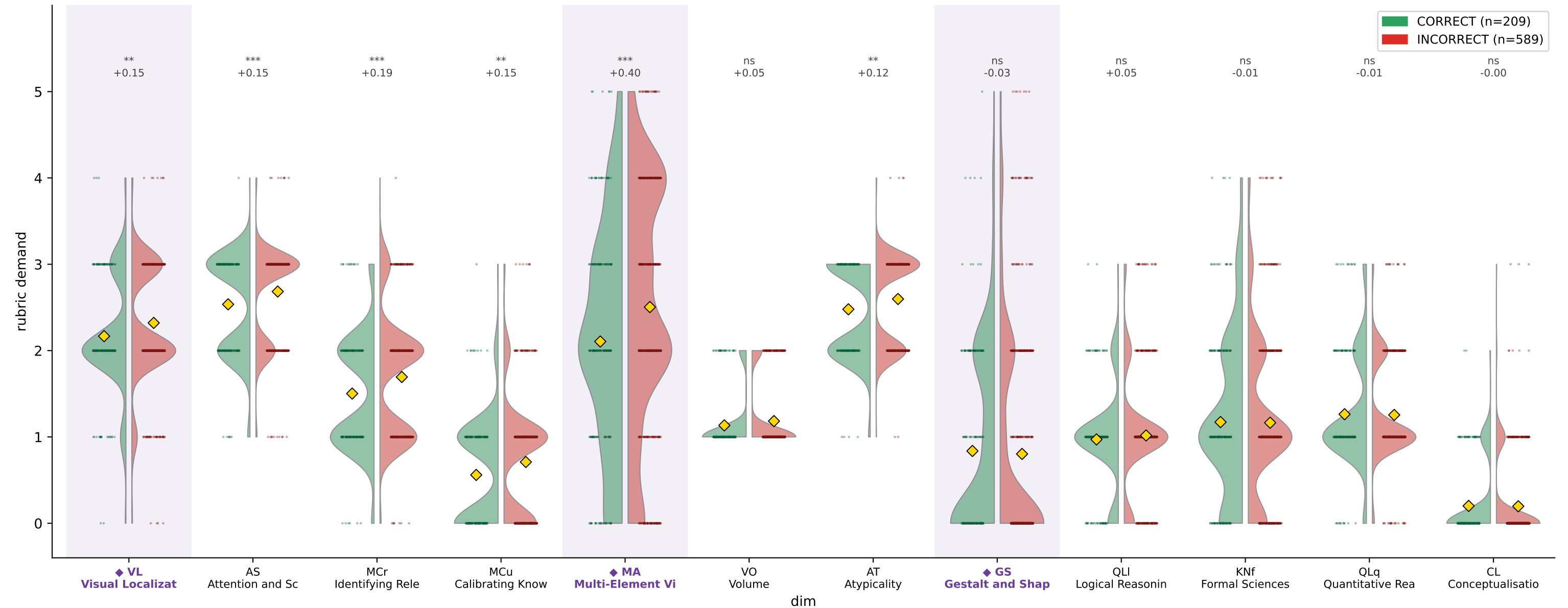
CharXiv DeLeAn demand by reasoning correctness — Ovis1.5-Gemma2-9B (train)



CharXiv DeLeAn demand by reasoning correctness — Cambrian-34B (train)

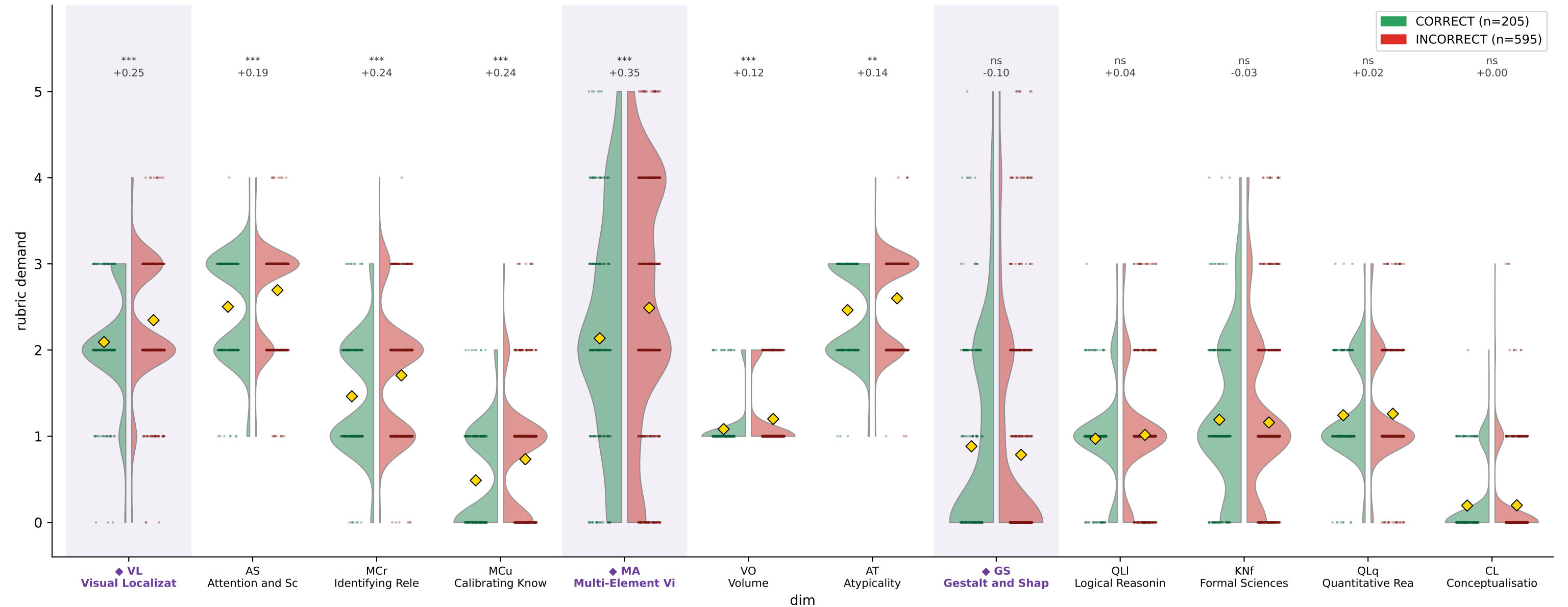


CharXiv DeLeAn demand by reasoning correctness — Reka-Flash (train)

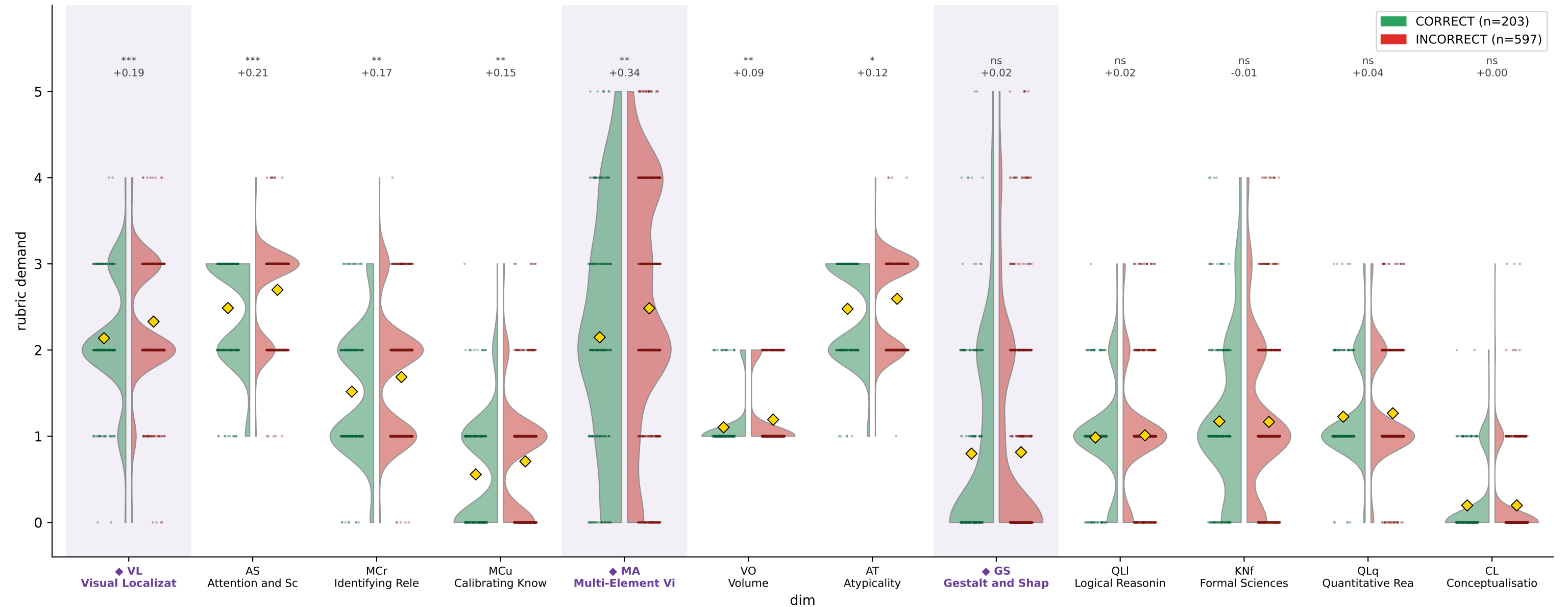


Violins = demand distribution (KDE); jittered points = items (≤ 600 /group shown); $\blacklozenge$ diamond = mean. Δ = incorrect – correct (positive $\Rightarrow$ failed items higher-demand). Mann-Whitney U: *** $p < .001$, ** $p < .01$, * $p < .05$, ns. Dims in fixed global order; $\blacklozenge$ /purple = emergent

CharXiv DeLeAn demand by reasoning correctness — Qwen-VL-Max (train)

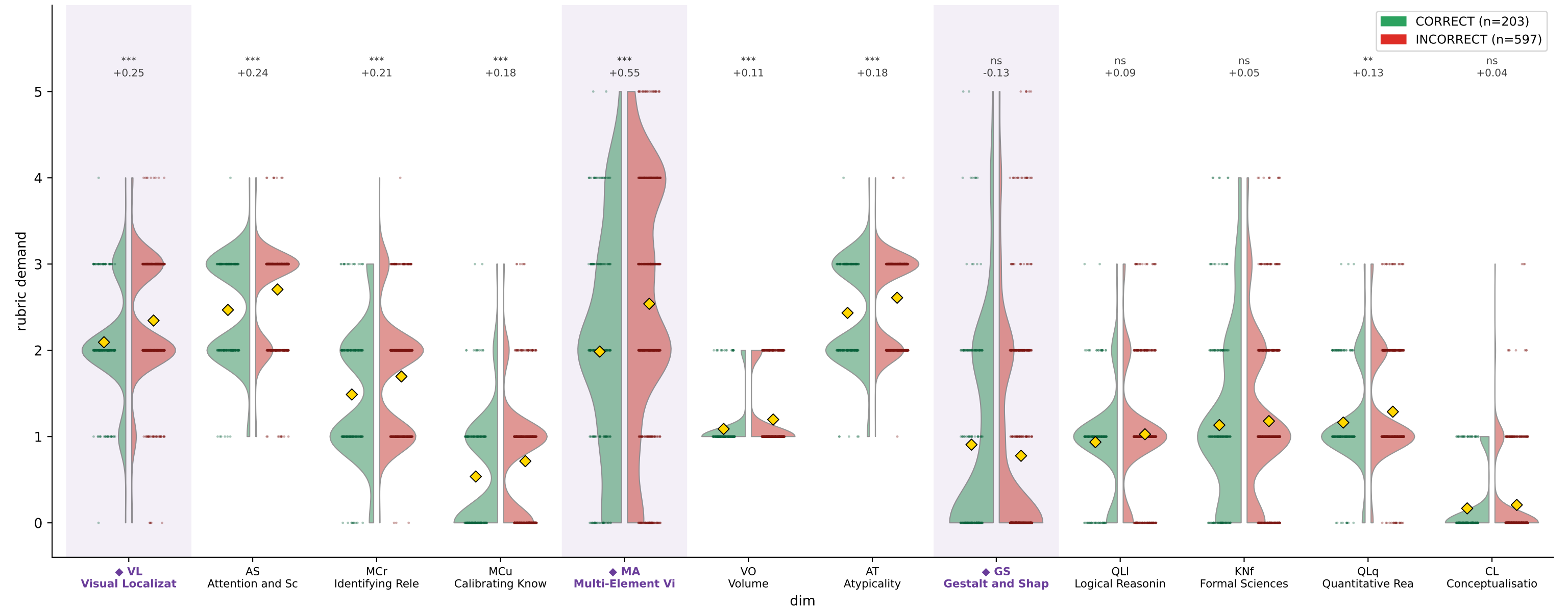


CharXiv DeLeAn demand by reasoning correctness — InternLM-XComposer2-4KHD (train)

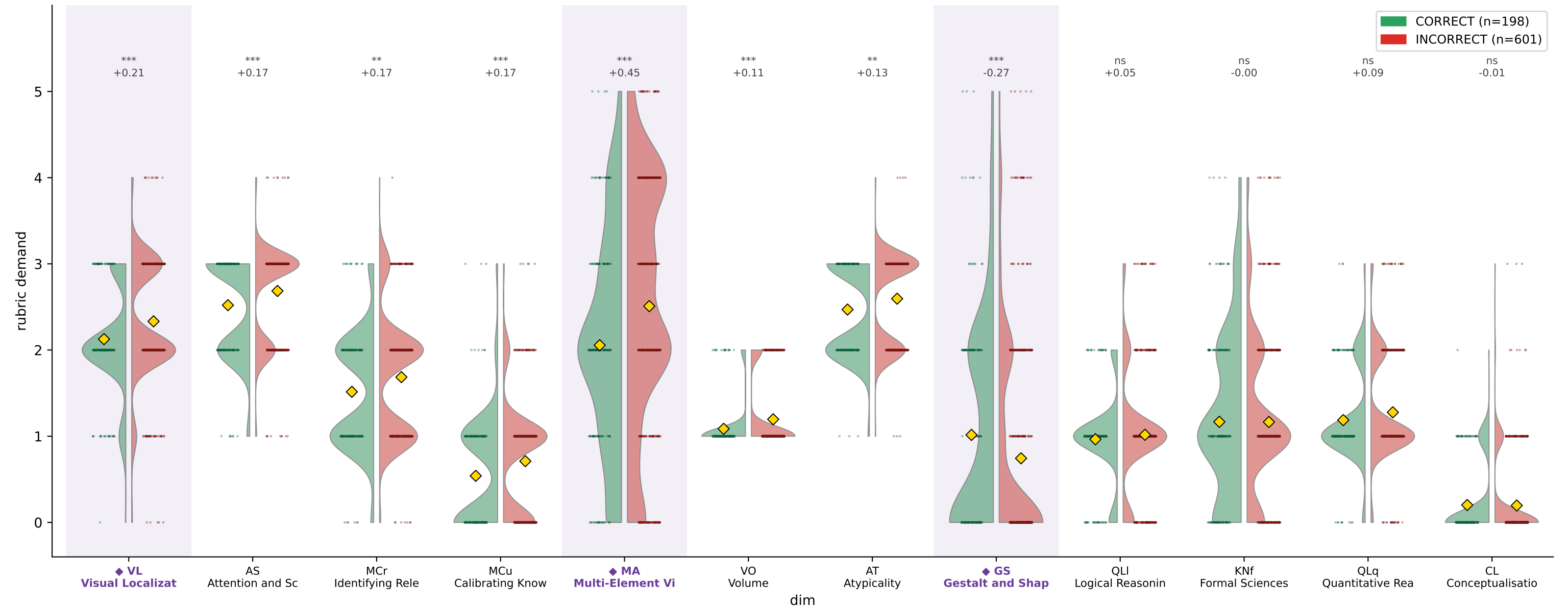


violins = demand distribution (KDE); jittered points = items (≤ 600 /group shown); ◆ diamond = mean. Δ = incorrect – correct (positive $\Rightarrow$ failed items higher-demand). Mann-Whitney U: *** $p < .001$, ** $p < .01$, * $p < .05$, ns. Dims in fixed global order; ◆/purple = emergent

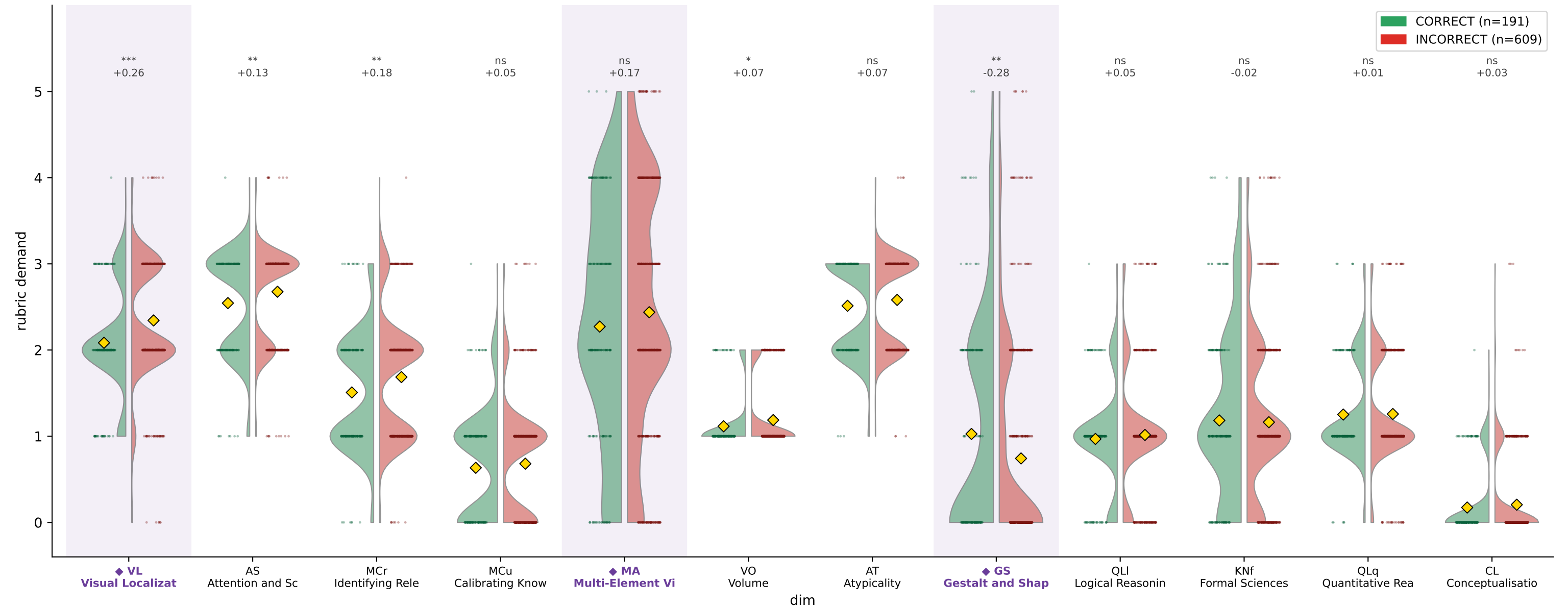
CharXiv DeLeAn demand by reasoning correctness — MiniCPM-V2-5 (train)



CharXiv DeLeAn demand by reasoning correctness — Mini-Gemini-HD-Yi-34B (train)

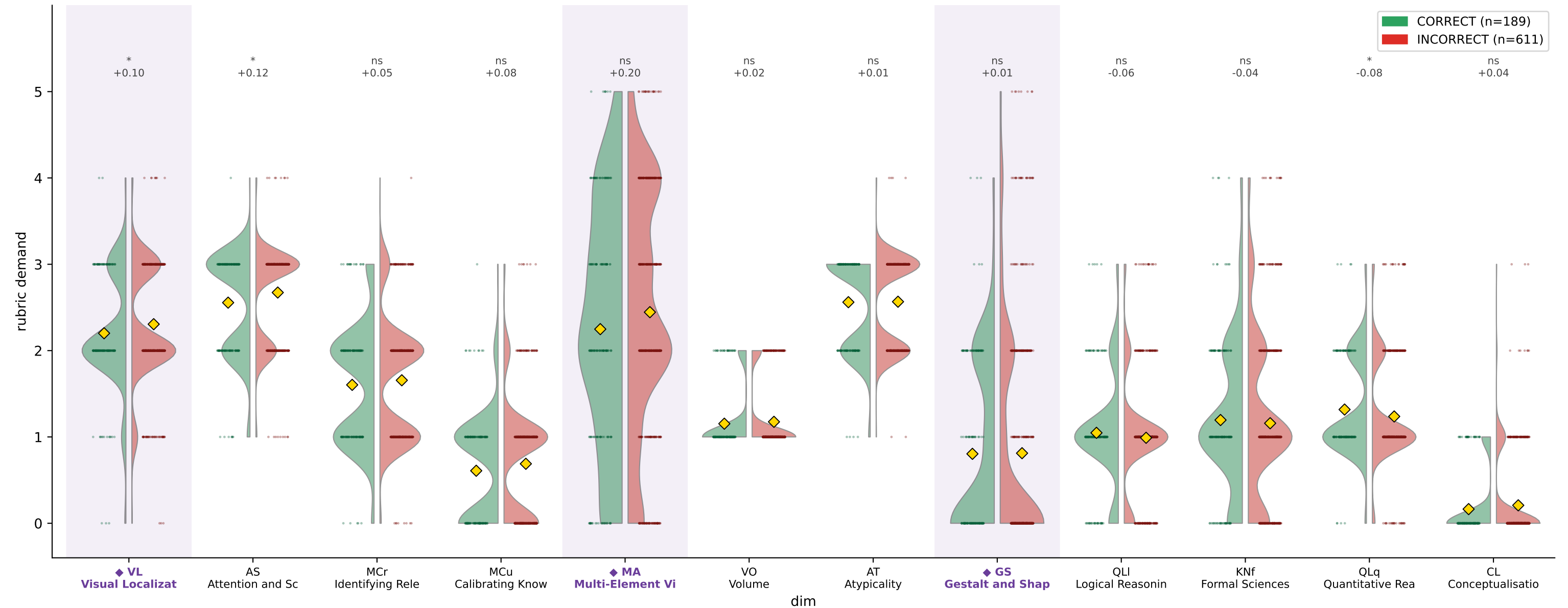


CharXiv DeLeAn demand by reasoning correctness — VILA-1-5-40B (train)



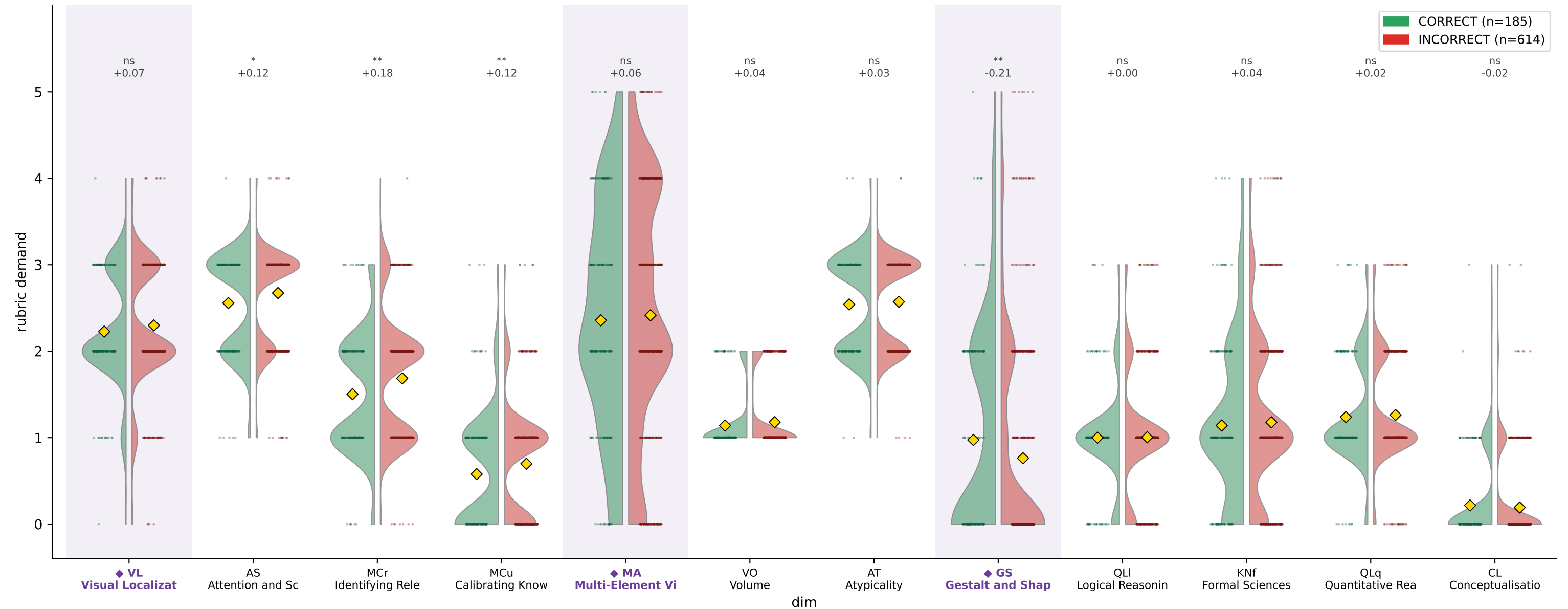
Violins = demand distribution (KDE); jittered points = items (≤ 600 /group shown); $\blacklozenge$ diamond = mean. Δ = incorrect – correct (positive $\Rightarrow$ failed items higher-demand). Mann-Whitney U: *** $p < .001$, ** $p < .01$, * $p < .05$, ns. Dims in fixed global order; $\blacklozenge$ /purple = emergent

CharXiv DeLeAn demand by reasoning correctness — Reka-Edge (train)

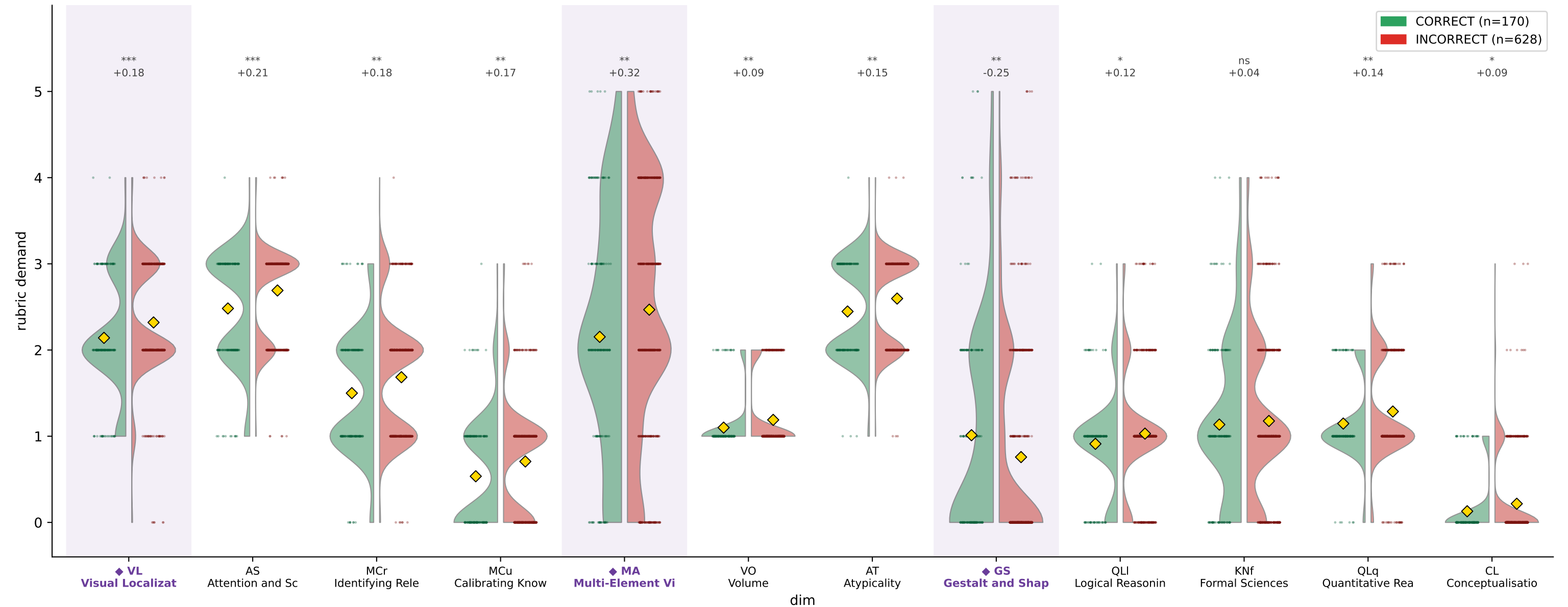


violins = demand distribution (KDE); jittered points = items (≤ 600 /group shown); $\blacklozenge$ diamond = mean. Δ = incorrect – correct (positive $\Rightarrow$ failed items higher-demand). Mann-Whitney U: *** $p < .001$, ** $p < .01$, * $p < .05$, ns. Dims in fixed global order; $\blacklozenge$ /purple = emergent

CharXiv DeLeAn demand by reasoning correctness — Gemini-1.0-Pro-Vision (train)

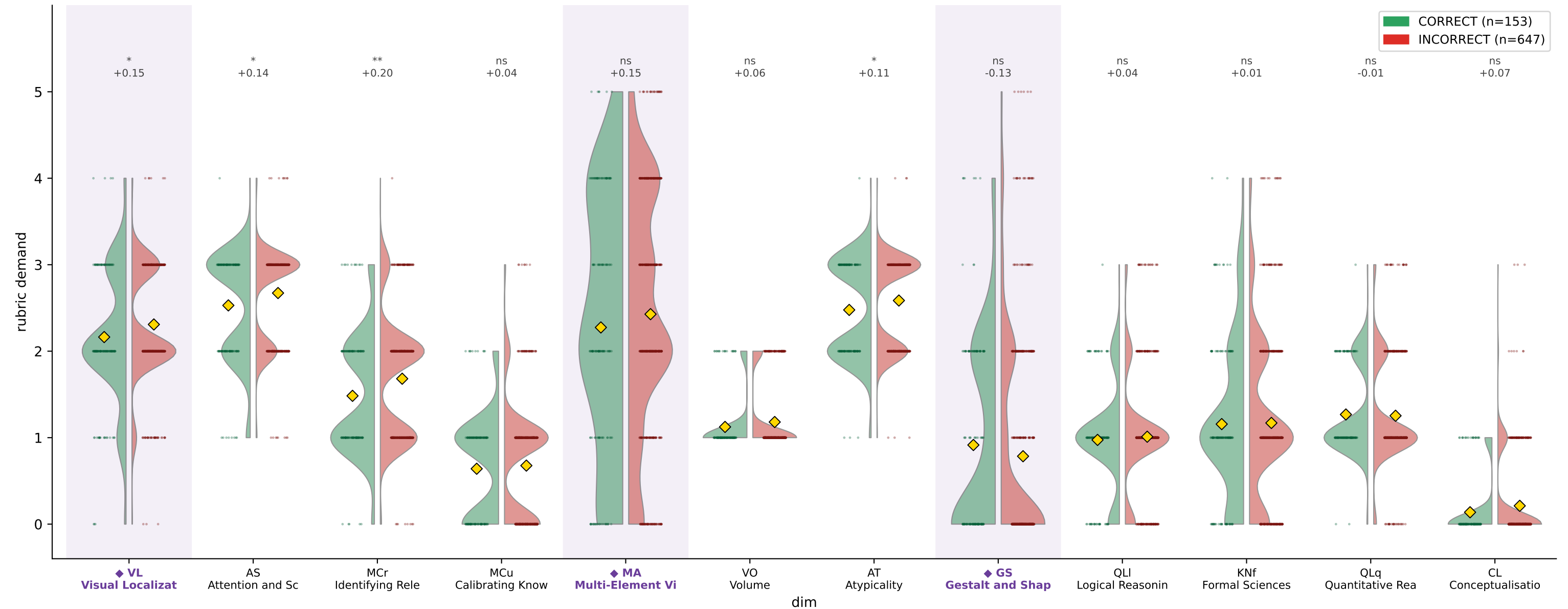


CharXiv DeLeAn demand by reasoning correctness — LLaVA-1-6-Yi-34B (train)

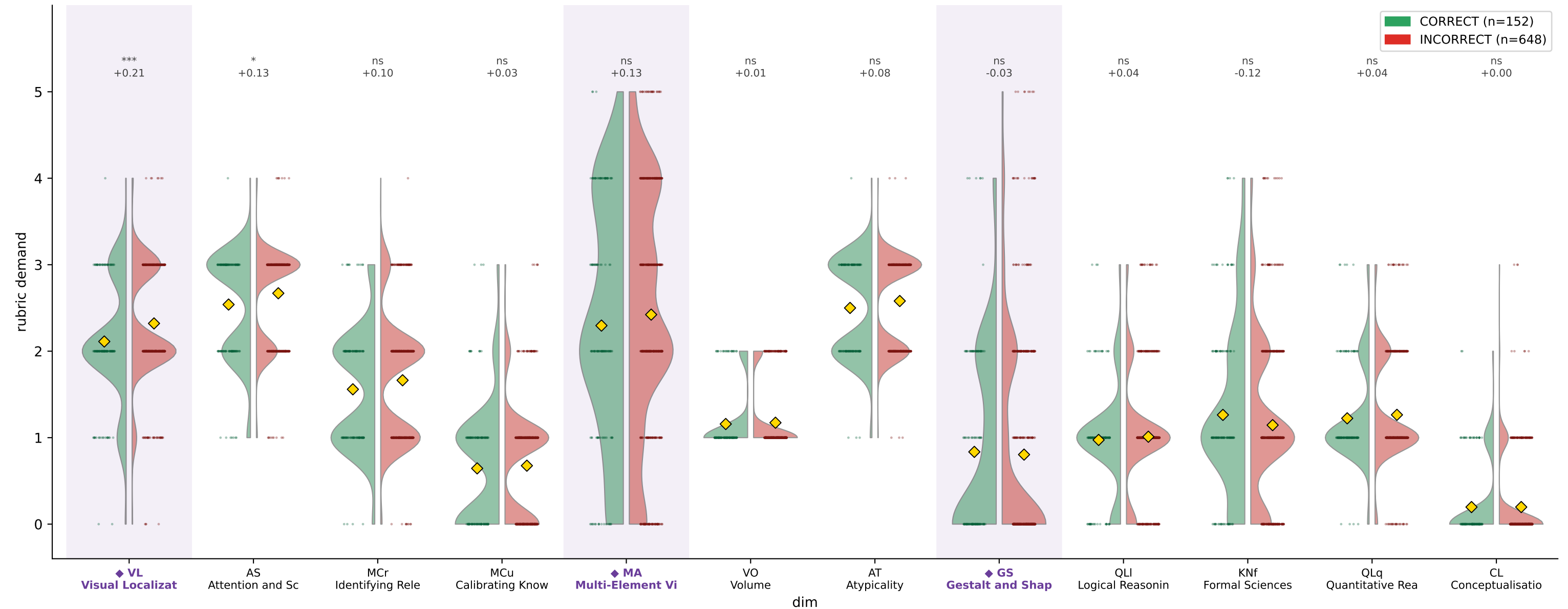


Violins = demand distribution (KDE); jittered points = items (≤ 600 /group shown); $\blacklozenge$ diamond = mean. Δ = incorrect – correct (positive $\Rightarrow$ failed items higher-demand). Mann-Whitney U: *** $p < .001$, ** $p < .01$, * $p < .05$, ns. Dims in fixed global order; $\blacklozenge$ /purple = emergent

CharXiv DeLeAn demand by reasoning correctness — MoAI (train)

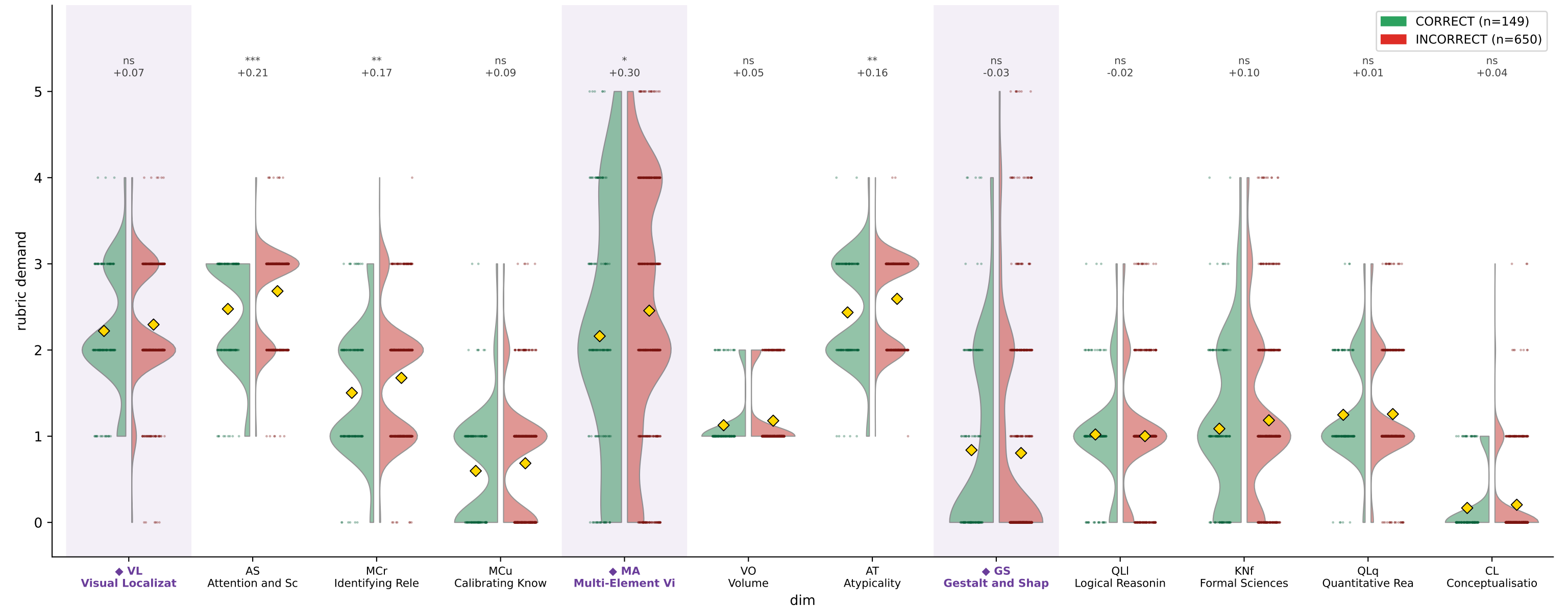


CharXiv DeLeAn demand by reasoning correctness — IDEFICS-2 (train)

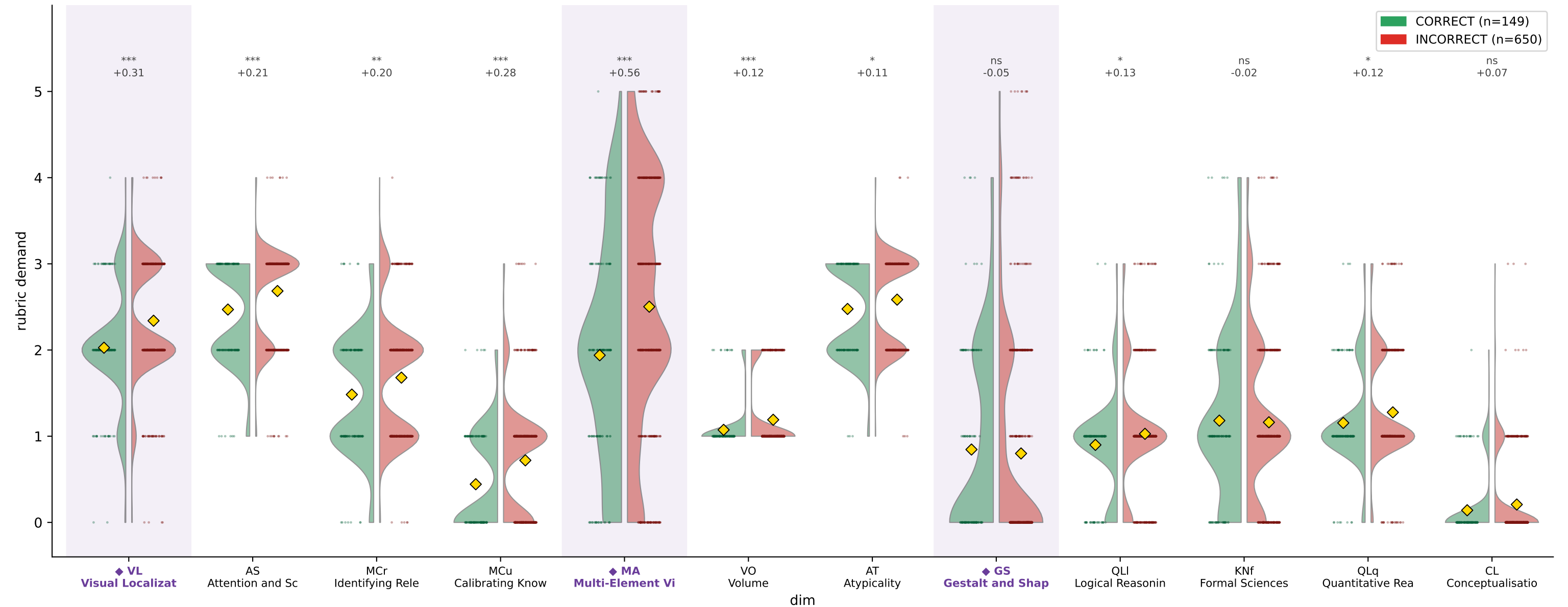


Violins = demand distribution (KDE); jittered points = items (≤ 600 /group shown); ♦ diamond = mean. Δ = incorrect – correct (positive $\Rightarrow$ failed items higher-demand). Mann-Whitney U: *** $p < .001$, ** $p < .01$, * $p < .05$, ns. Dims in fixed global order; ♦/purple = emergent

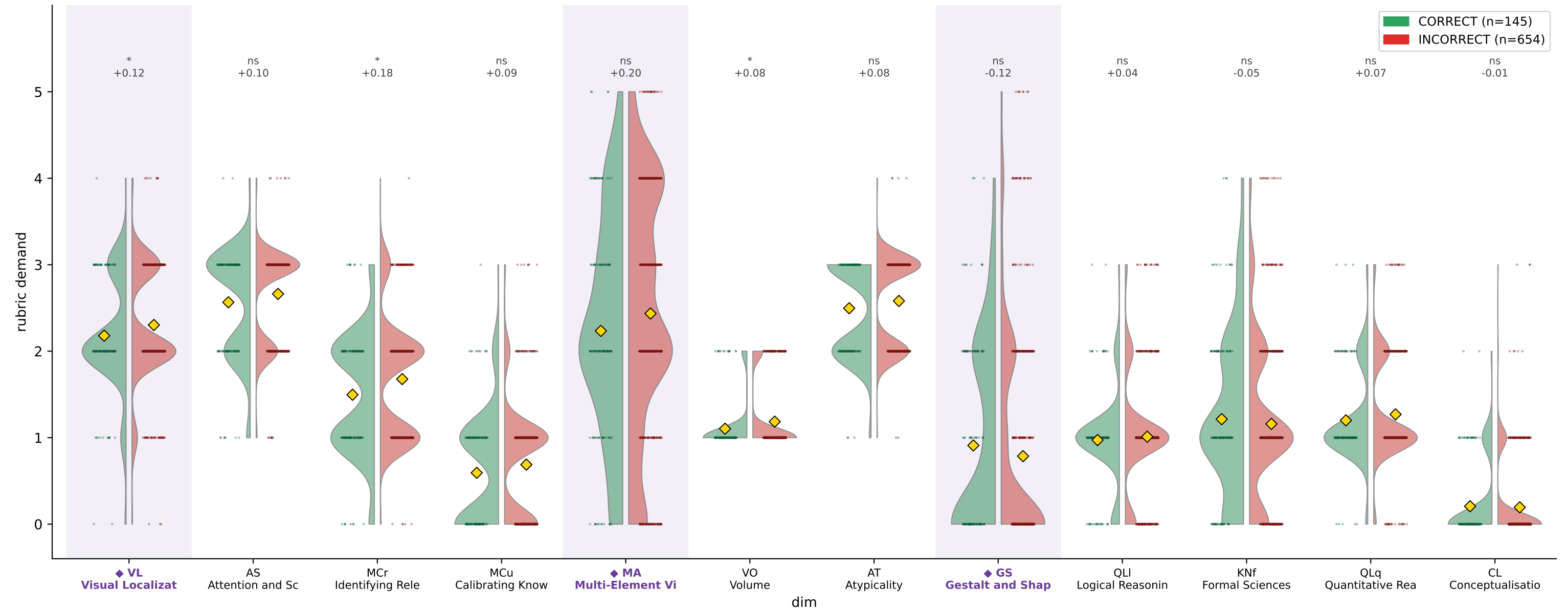
CharXiv DeLeAn demand by reasoning correctness — InternLM-XComposer2 (train)



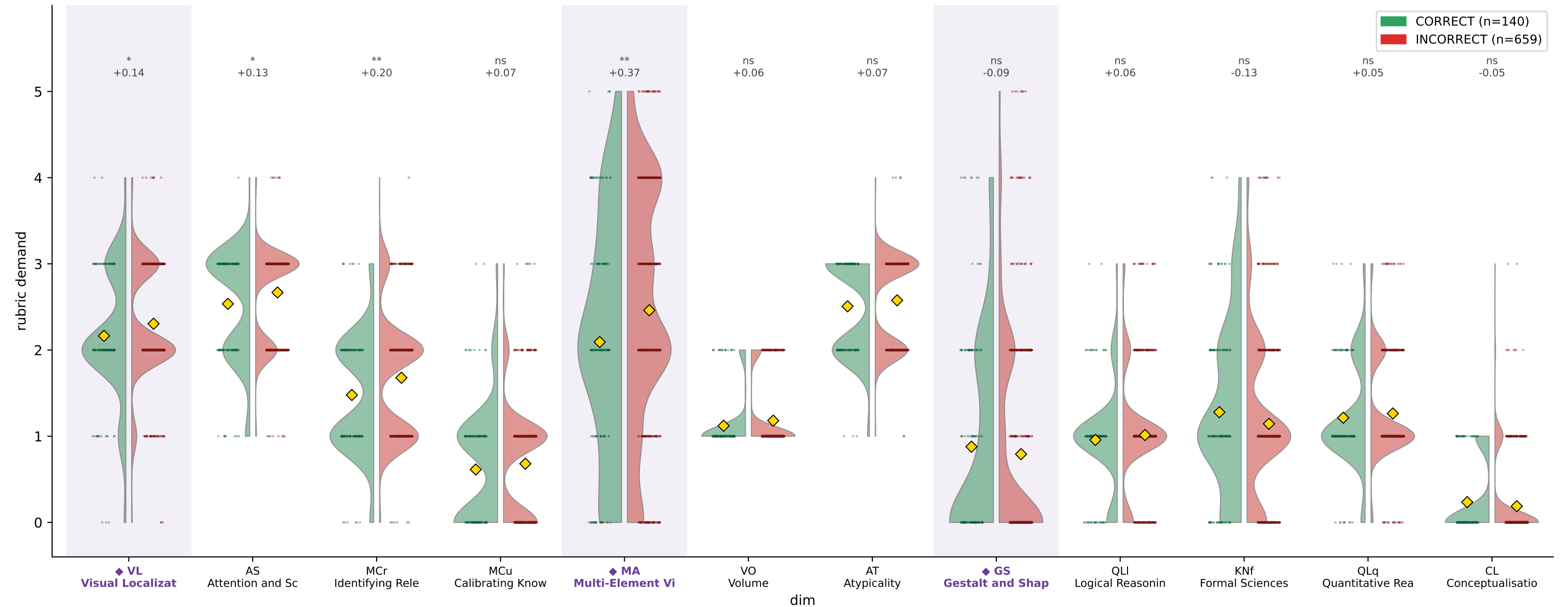
CharXiv DeLeAn demand by reasoning correctness — Mini-Gemini-HD-LLaMA3-8B (train)



CharXiv DeLeAn demand by reasoning correctness — MiniCPM-V2 (train)

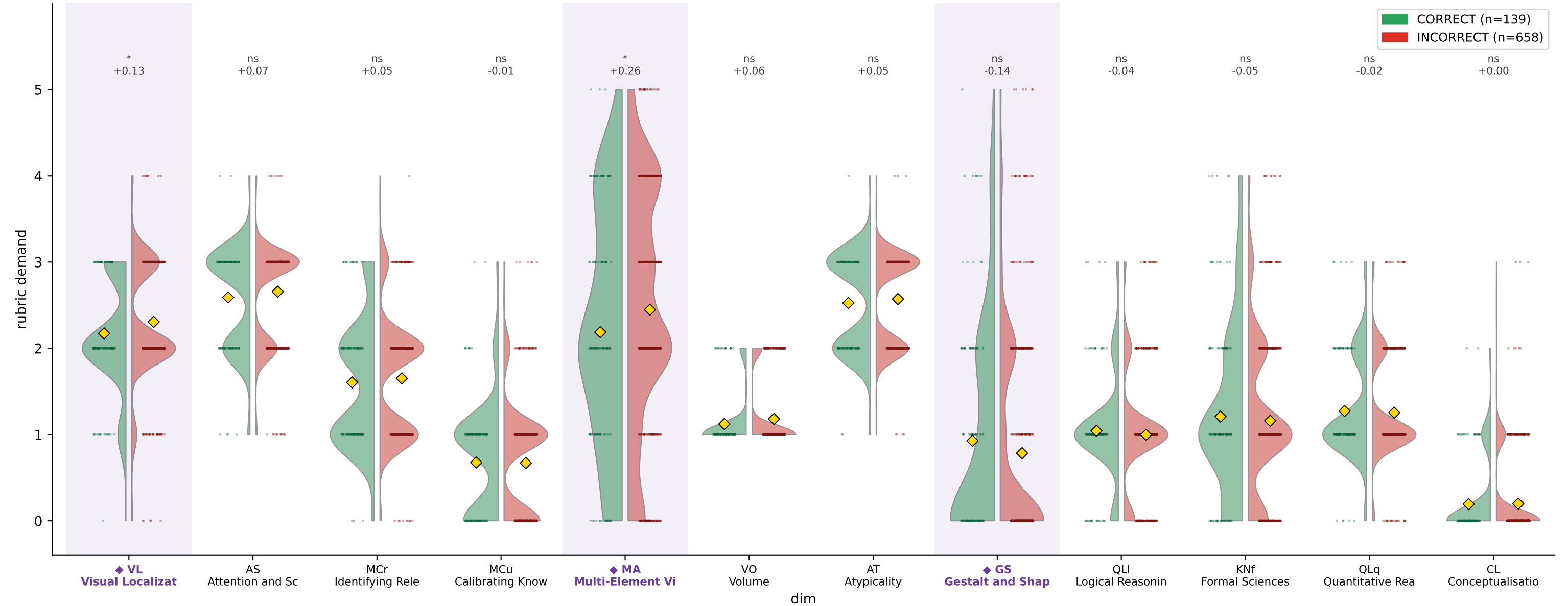


CharXiv DeLeAn demand by reasoning correctness — DeepSeek-VL (train)



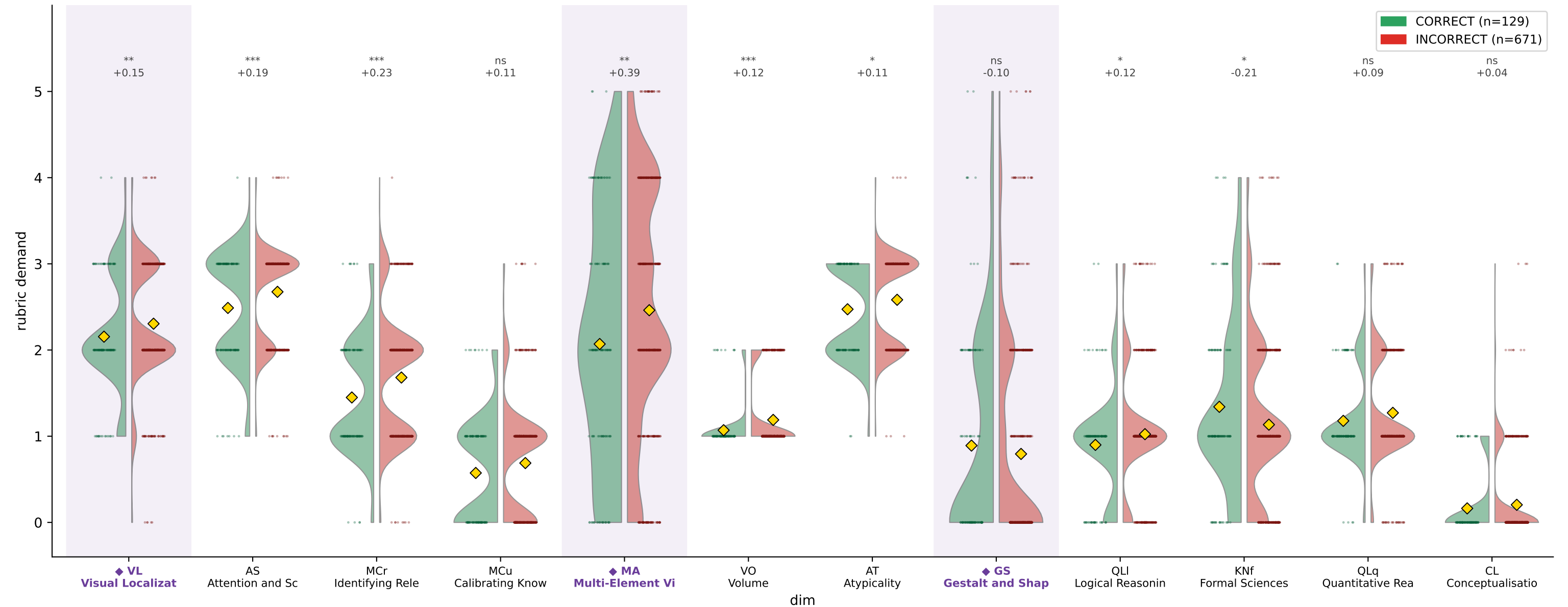
violins = demand distribution (KDE); jittered points = items (≤ 600 /group shown); ◆ diamond = mean. Δ = incorrect – correct (positive $\Rightarrow$ failed items higher-demand). Mann-Whitney U: *** $p < .001$, ** $p < .01$, * $p < .05$, ns. Dims in fixed global order; ◆/purple = emergent

CharXiv DeLeAn demand by reasoning correctness — IDEFICS-2-Chatty (train)

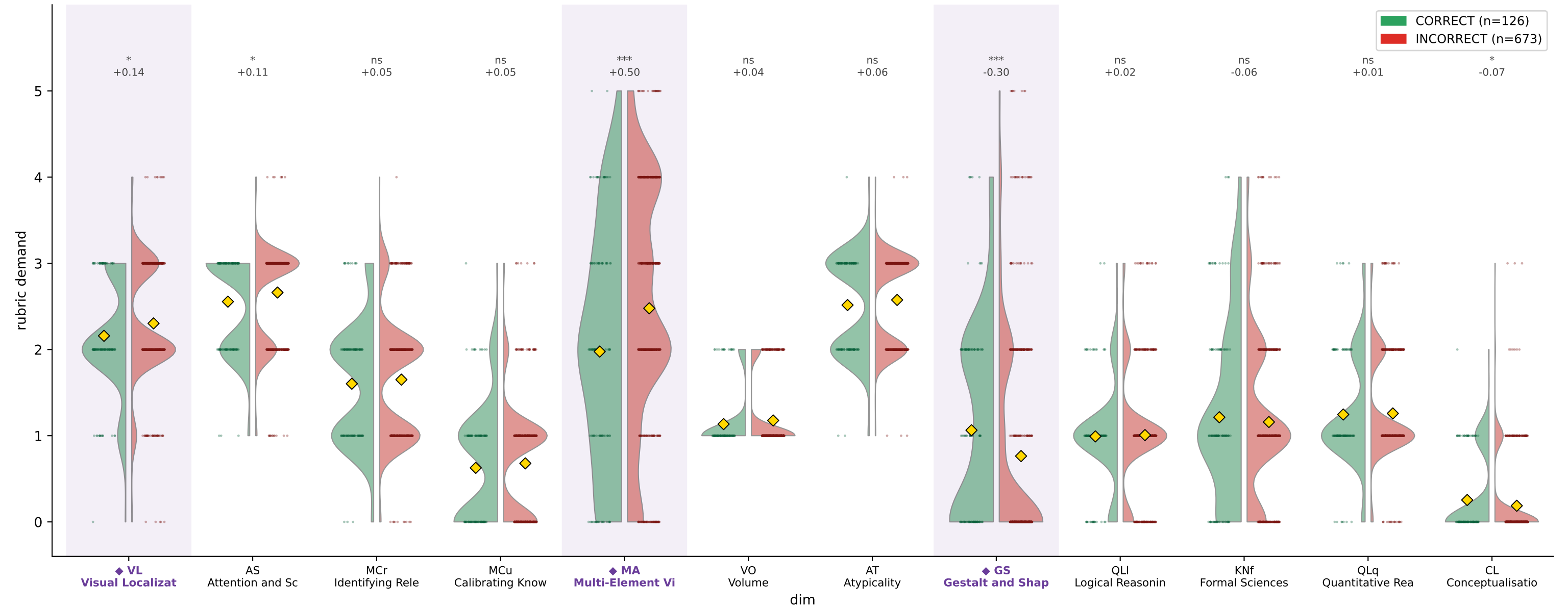


violins = demand distribution (KDE); jittered points = items (≤ 600 /group shown); $\blacklozenge$ diamond = mean. Δ = incorrect – correct (positive $\Rightarrow$ failed items higher-demand). Mann-Whitney U: *** $p < .001$, ** $p < .01$, * $p < .05$, ns. Dims in fixed global order; $\blacklozenge$ /purple = emergent

CharXiv DeLeAn demand by reasoning correctness — SPHINX-V2-13B (train)

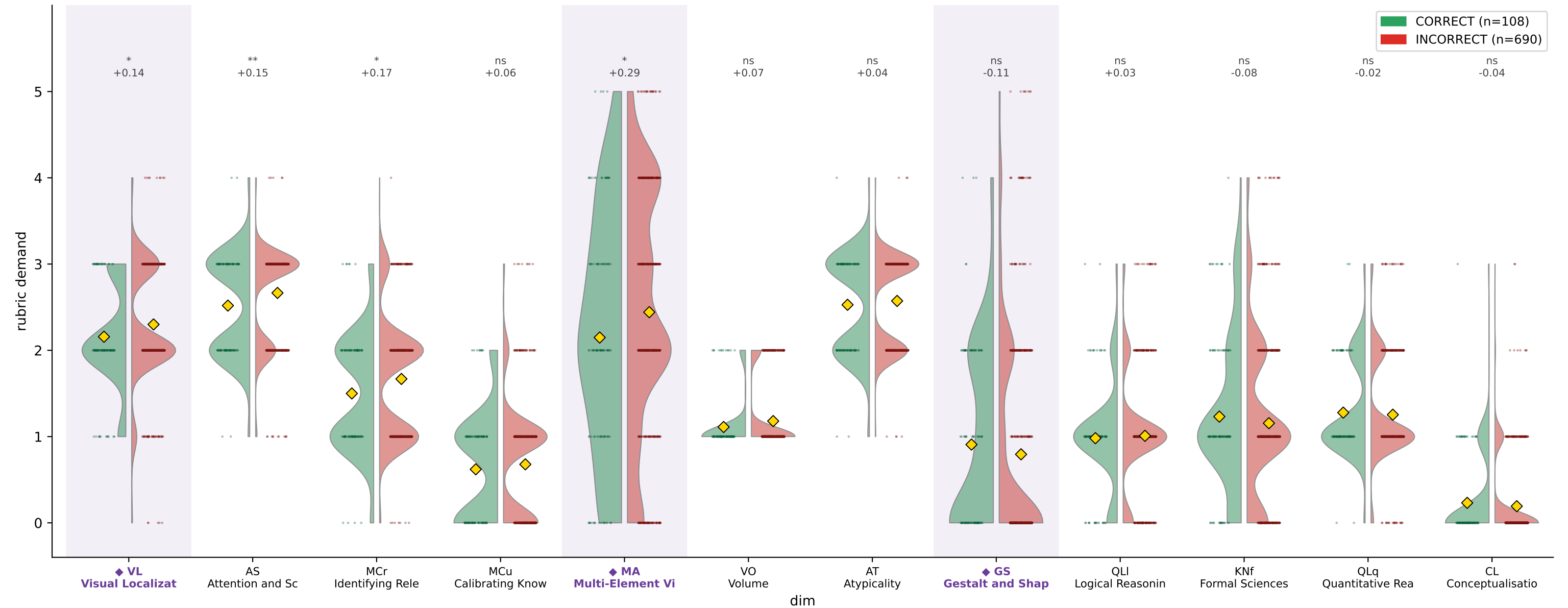


CharXiv DeLeAn demand by reasoning correctness — Qwen-VL-Plus (train)

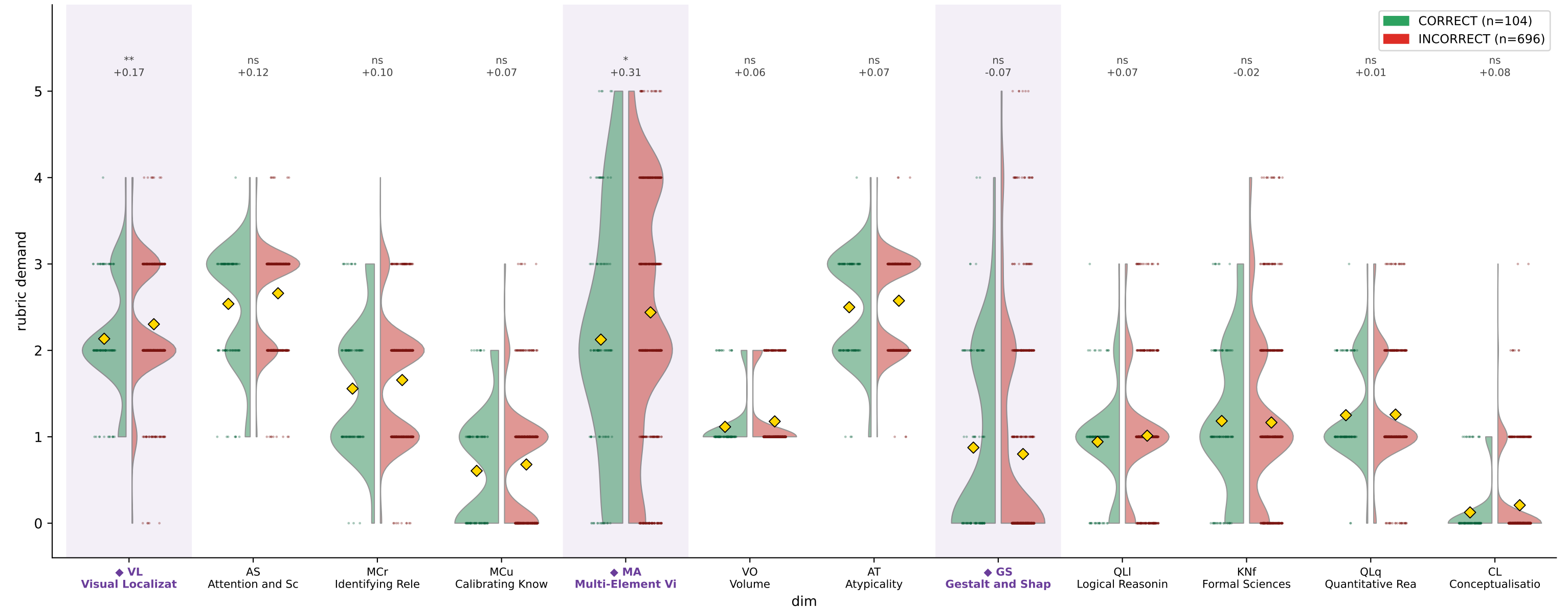


violins = demand distribution (KDE); jittered points = items (≤ 600 /group shown); ♦ diamond = mean. Δ = incorrect – correct (positive $\Rightarrow$ failed items higher-demand). Mann-Whitney U: *** $p < .001$, ** $p < .01$, * $p < .05$, ns. Dims in fixed global order; ♦/purple = emergent

CharXiv DeLeAn demand by reasoning correctness — LLaVA-1-6-Mistral-7B (train)

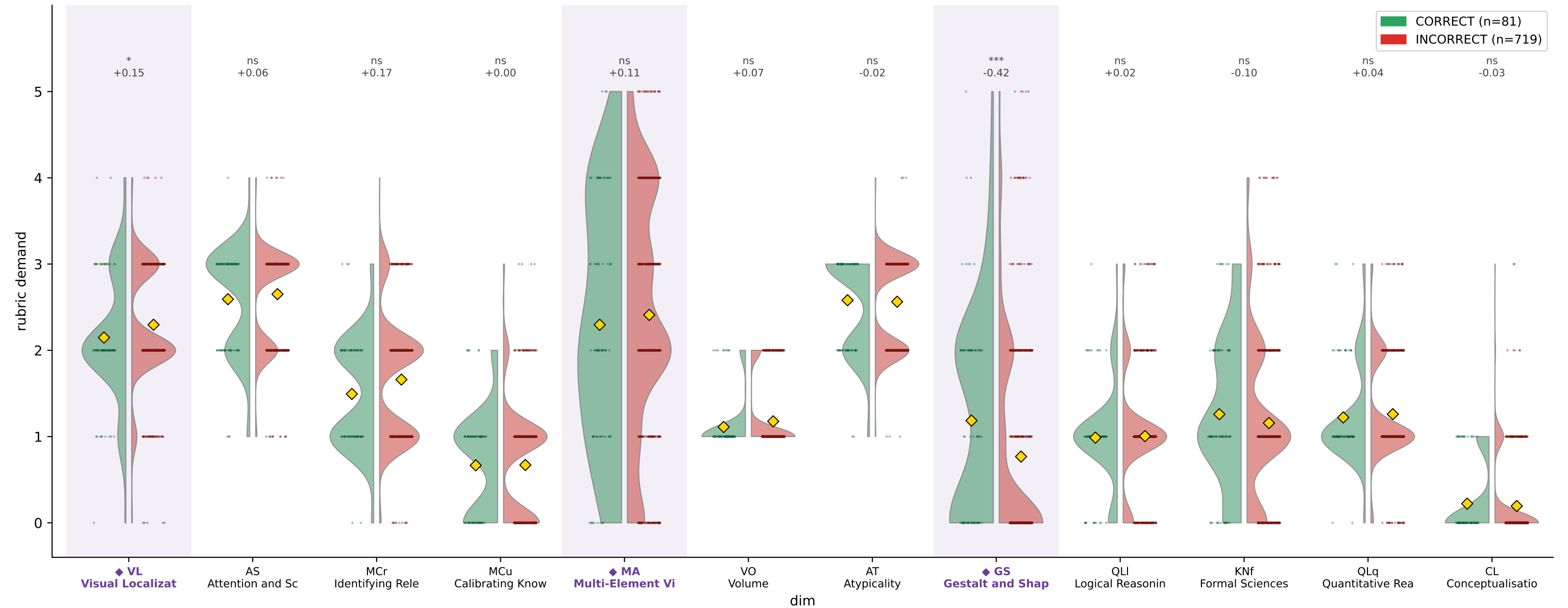


CharXiv DeLeAn demand by reasoning correctness — ChartGemma (train)



Violins = demand distribution (KDE); jittered points = items (≤ 600 /group shown); $\blacklozenge$ diamond = mean. Δ = incorrect – correct (positive $\Rightarrow$ failed items higher-demand). Mann-Whitney U: *** $p < .001$, ** $p < .01$, * $p < .05$, ns. Dims in fixed global order; $\blacklozenge$ /purple = emergent

CharXiv DeLeAn demand by reasoning correctness — GPT-4o-Random (train)



Violins = demand distribution (KDE); jittered points = items (≤ 600 /group shown); $\blacklozenge$ diamond = mean. Δ = incorrect – correct (positive $\Rightarrow$ failed items higher-demand). Mann-Whitney U: *** $p < .001$, ** $p < .01$, * $p < .05$, ns. Dims in fixed global order; $\blacklozenge$ /purple = emergent